

Probabilistic Race and Ethnicity Prediction Using Group-Specific Name Lists

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Abstract

Statistically valid estimation of racial and ethnic disparities often requires inferring the probability that an individual belongs to a particular racial or ethnic group, given only their name and geographic location. The standard approach, Bayesian Improved Surname Geocoding (BISG), relies on group population frequencies for each name. Although the U.S. Census Bureau provides such information for common names and a limited set of racial categories, comparable data do not exist for many racial and ethnic groups and are rarely available outside the U.S. We propose the list-powered BISG (ℓ BISG) method, which can be used to derive calibrated group probabilities from group-specific name lists. These lists may be compiled based on expert knowledge or generated synthetically using large language models (LLMs), and thus may be subject to unknown biases. Representing names as embeddings, we treat list membership as a proxy prediction task and apply a correction based on proximal inference to recover the target group probabilities. We validate the method on U.S. voter files with self-reported race, on the full-count 1900 U.S. Census, and on the Lebanese voter registry. We find that LLM-generated name lists yield accurate and well-calibrated probabilities as well as precise disparity estimates comparable to those obtained using methods that require name–race data. Thus, ℓ BISG substantially broadens the applicability of probabilistic race and ethnicity prediction to settings where name–race data are unavailable.

Keywords: Bayesian Improved Surname Geocoding, embeddings, large language models, proximal inference, racial disparity

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1 Introduction

Measuring racial and ethnic disparities across social, economic, and political domains requires information about individuals’ race in administrative and other records (e.g., [Adjaye-Gbewonyo et al., 2014](#); [Fraga, 2018](#); [Greenwald et al., 2024](#)). However, race is often unavailable because it is withheld by law or omitted due to privacy concerns and the sensitivity of collecting such information ([Fiscella and Fremont, 2006](#)). As a result, researchers must infer individuals’ racial or ethnic group membership from the information that is available, most commonly their name and place of residence.

Bayesian Improved Surname Geocoding (BISG) combines surname-based racial information, derived from the U.S. Census Bureau’s surname list, with the racial composition of an individual’s place of residence to estimate individual race probabilities. The method assumes that surname and geolocation are conditionally independent given unobserved race ([Elliott et al., 2009](#); [Imai and Khanna, 2016](#); [Imai et al., 2022](#)). BISG has been extended to incorporate first names, resulting in the Bayesian Improved First Name Surname Geocoding (BIFSG) method ([Voicu, 2018](#)).

BISG has become the standard approach for racial demographic inference across many domains. Its applications range from fair lending enforcement ([Consumer Financial Protection Bureau, 2014](#); [Greenwald et al., 2024](#)) to studies of health disparities ([Adjaye-Gbewonyo et al., 2014](#)) and racial turnout gaps in voting ([Fraga, 2018](#)), and it is supported by widely used software implementations ([Khanna et al., 2024](#)). BISG returns a probability for each group rather than a single label, which is essential for downstream statistical analyses. Probabilistic outputs enable unbiased estimation of group disparities in settings where deterministic classifications can introduce bias ([McCartan et al., 2025](#)).

However, BISG requires data that exist only in some settings. It requires a surname-by-race frequency table (and BIFSG additionally requires an analogous table for first names), as well as the geographic prevalence of each racial group. The U.S. Census Bureau publishes these tables only for names occurring at least 100 times and for a limited set of racial categories ([U.S. Census Bureau, 2016](#)). Consequently, individuals with uncommon names receive only an uninformative national prior. The required name-frequency tables are unavailable for more detailed ethnic groups and for most countries outside the United States.

We propose list-powered BISG (ℓ BISG), which estimates calibrated race probabilities without requiring a name-by-race frequency table (Section 2). We consider two settings: one in which geographic prevalence is available for all groups of interest and another in which it is available only for a coarser group (Section 3). The proposed ℓ BISG relies on relatively short, approximate group-specific name lists, which can be compiled from expert knowledge or generated synthetically using large language models (LLMs) (Section 4). To leverage these lists, we first embed names into a vector space, in which morphologically and linguistically similar names are close to one another ([Ye et al., 2017](#); [Wang et al., 2024](#); [Dasanaïke, 2026](#); [Dasanaïke and Imai, 2026](#)). We then apply a proximal inference approach ([Kuroki and Pearl, 2014](#)) to recover race probabilities from a model that predicts whether a name embedding belongs to a given list.

Like standard BISG ([Voicu, 2018](#)), ℓ BISG naturally extends to incorporate group-specific first name lists, yielding ℓ BIFSG. Because it produces calibrated race probabilities, ℓ BISG can also substitute for BISG in downstream methods such as BIRDIE ([McCartan et al., 2025](#)),

enabling valid estimation of group disparities without modification.

We evaluate the empirical performance of ℓ BISG using three datasets with individual-level ground truth (Section 5): (1) the Florida and North Carolina voter files, which include self-reported race for registered voters; (2) the complete-count 1900 U.S. Census, which enumerates Chinese and Japanese as separate races (Nelson et al., 2025); and (3) the 2018 Lebanese voter file, scraped from government records, containing individuals’ self-reported religious sect. Across all three settings, ℓ BISG achieves predictive accuracy and probability calibration comparable to those of BISG. We also show that this strong performance carries over to downstream analyses, yielding disparity estimates that are nearly identical to those obtained using BISG under the BIRDIE framework (McCartan et al., 2025).

By eliminating the need for name–race frequency tables and requiring only easily constructed name lists, ℓ BISG substantially broadens the scope of race and ethnicity imputation.

Related literature. A growing literature seeks to overcome the limitations of the Census surname list by training machine-learning models to infer race or ethnicity directly from names. Character-level and transformer-based models predict race from the sequence of letters in a name (Sood and Laohaprapanon, 2018; Parasurama, 2021); interpretable models leverage patterns in name popularity (Jain et al., 2022); and embedding-based approaches learn representations of names from social and communication data without using race labels (Ye et al., 2017; Ye and Skiena, 2019). More recently, LLMs have been used to infer ethnicity, religious sect, or caste from names without task-specific training, both in the United States and elsewhere (Dasanaïke, 2026).

These approaches typically assign each individual to a single group and are often proposed as alternatives to BISG. However, classification errors make naïvely treating these predictions as ground truth problematic, leading to biased estimates in downstream disparity analyses (Chen et al., 2019; Xin et al., 2026). Rather than replacing BISG, ℓ BISG leverages the outputs of these and other classifiers to recover calibrated probabilistic predictions, which can then be used for statistically valid downstream inference (McCartan et al., 2025).

A number of existing methods extend BISG itself. As noted above, Voicu (2018) incorporates first-name information through an additional Bayes factor (BIFSG), while Imai et al. (2022) develops a fully Bayesian version (fBISG) that accounts for potential measurement error in the Census surname data. Rosenman et al. (2023) augments the Census surname list with name dictionaries derived from Southern voter files, and Dasanaïke and Imai (2026) replaces the uninformative prior for uncommon surnames with embedding-based predictions (eBISG). Lastly, Greengard and Gelman (2025) improves the calibration of BISG probabilities to relax the surname-geography independence assumption. Nevertheless, all of these methods rely on a name–race frequency table. In contrast, ℓ BISG eliminates this requirement while remaining compatible with the same downstream analyses and naturally extending to incorporate first-name information (Sections 2.6 and 5.1).

Curated name lists have long been used to identify and sample members of specific populations. Distinctive surname lists have been developed for American Jews (Himmelfarb et al., 1983), Asian Americans (Lauderdale and Kestenbaum, 2000), and, with formal validation, South Asian and Chinese populations in administrative health data (Shah et al., 2010). These approaches often use curated lists as deterministic filters, classifying individuals solely on the basis of whether their names appear on the list (see Chasalow et al.

(2026) for an exception). In contrast, ℓ BISG treats a list as a noisy proxy and generalizes beyond the listed names using name embeddings, allowing previously unseen names to receive informative probabilistic predictions.

Finally, name-based prediction is increasingly used to measure representation. Recent applications include inferring the race of public officials (Lee and Velez, 2025) and evaluating machine-learning variants of BISG for redistricting (DeLuca and Curiel, 2024). At the same time, prediction errors have been shown to distort the downstream analyses that rely on these predictions (Argyle and Barber, 2024; Greenwald et al., 2024). By making calibrated probabilistic predictions available for a much broader range of groups and settings, ℓ BISG extends the scope of these applications while supporting statistically valid downstream inference.

2 The List-powered BISG Method

We now introduce our ℓ BISG methodology for obtaining the necessary probabilistic predictions when information about the frequency of names by group is unavailable. In this section, we assume that information about the geographic distribution of the group variable R is available, meaning that $\mathbb{P}(R = r \mid G = g)$ are known from census data or have been estimated from surveys (see Section 3 for an extension to the setting, in which such information is available only for a coarser group).

However, for surnames, we only suppose that for each value of R , we have a list of names that are distinctive, common, or typical of that group. These lists may even have some names in common, though more overlap will make groups less distinguishable. In some cases, one group may represent a residual or “other” group not of interest for analysis. Although helpful to have a list for this group if possible, it will not be strictly necessary for ℓ BISG. In Section 4, we further discuss desirable properties of such lists and how one may obtain them. For now, let $L_r = L_r(S)$ be a binary indicator for group r which is 1 if name S is in that group’s list and 0 otherwise.

2.1 The value of probabilistic prediction in downstream analysis

Before introducing ℓ BISG, we briefly explain why probabilistic prediction of an individual’s race is essential for valid statistical estimation of racial disparities. Suppose we observe a random sample of n individuals from a population. For each individual, we observe their geographic location $G_i \in \mathcal{G}$, name $S_i \in \mathcal{S}$ (a first name, surname, or both), and outcome $Y_i \in \mathcal{Y}$, but not their membership $R_i \in \mathcal{R}$ in a racial, ethnic, or other group of interest. Let $d_{\mathcal{G}} := |\mathcal{G}|$, $d_{\mathcal{S}} := |\mathcal{S}|$, and $d_{\mathcal{R}} := |\mathcal{Y}|$ denote the numbers of possible values of these variables in the population, and assume each is finite. Our objective is to predict R_i from each individual’s name S_i and geographic location G_i , and then use these probabilistic predictions to estimate quantities such as disparities in the outcome Y across groups.

A common approach is to assign each individual to the group with the highest predicted probability based on their name and geographic location. However, this hard classification ignores classification error and can bias downstream analyses (Chen et al., 2019; Xin et al., 2026). To illustrate, consider the binary case in which both R and Y are binary. Suppose

each individual is assigned to the racial group with the highest probability, i.e.,

$$R^* = \arg \max_{r \in \{0,1\}} \mathbb{P}(R = r \mid S, G).$$

We refer to this as the oracle classifier because it assumes the true conditional probabilities are known, thereby eliminating estimation error and model misspecification.

Now, suppose the outcome is independent of name conditional on race and geography, $Y \perp S \mid R, G$, which is the key identification assumption for downstream analyses of racial disparities (see [McCartan et al., 2025](#)). If we estimate the racial disparity in geography $G = g$ using the oracle classification R^* , the resulting bias is given by,

$$\begin{aligned} & [\mathbb{P}(Y = y \mid R^* = 1, g) - \mathbb{P}(Y = y \mid R^* = 0, g)] - [\mathbb{P}(Y = y \mid R = 1, g) - \mathbb{P}(Y = y \mid R = 0, g)] \\ &= - [\mathbb{P}(Y = y \mid R = 1, g) - \mathbb{P}(Y = y \mid R = 0, g)] \times [\mathbb{P}(R = 1 \mid S \in \mathcal{C}_{0g}, g) + \mathbb{P}(R = 0 \mid S \in \mathcal{C}_{1g}, g)], \end{aligned}$$

where $\mathcal{C}_{rg}$ denotes the set of names classified as $R^* = r$ in geography g ([Chen et al., 2019](#); [Xin et al., 2026](#)).

The bias in each geography is therefore proportional to the total misclassification rate and has the opposite sign of the true racial disparity. Consequently, hard classification systematically attenuates the estimated disparity toward zero: if the true racial disparity is positive, it is underestimated, whereas if it is negative, it is overestimated. Moreover, the bias is zero if and only if the classification is perfect or the true racial disparity is zero.

Empirically, we find that the misclassification rate can be sizable. For a brief illustration, we use the Florida voter file from Section 5.1, which contains self-reported race. We compute standard BISG probabilities $\mathbb{P}(R \mid S, G)$ for every voter using the Census surname table and tract composition. We estimate the racial composition of the electorate by assigning each registrant to their single most likely race and summing across individuals. Compared to the self-reported population composition, the hard classification counts over-predict the White share by about ten percentage points (ppts) and under-predict the Black share by about six ppts. Summing the estimated probabilities instead of counting the hard classifications reduces the mean absolute error of the estimated racial composition, averaged over the five categories, from 4.1 to 2.9 percentage points.

For downstream estimation, [McCartan et al. \(2025\)](#) proposes the BIRDIE framework, showing that if the outcome is conditionally independent of name given race and geography (i.e., $Y \perp\!\!\!\perp S \mid R, G$), racial disparities can be estimated without bias using calibrated racial probabilities, $\mathbb{P}(R = r \mid S, G)$, rather than imputing race through classification. This assumption is often more credible than the assumption required by the standard weighting approach, which assumes that race is conditionally independent of the outcome given name and geography (i.e., $Y \perp\!\!\!\perp R \mid S, G$) (see [McCartan et al., 2025](#), for details). Thus, accurate estimation of calibrated race probabilities is essential for racial disparity analysis. In our empirical validations (Section 5), we use this BIRDIE approach with ℓ BISG probabilities.

2.2 Using name embeddings for proxy prediction

Name lists contain information about the underlying patterns and structures associated with names from a particular group. For example, lists could suggest that surnames ending in “dez” are often Hispanic, whereas names containing “zh” are more likely to be Chinese.

Such regularities imply that names not appearing on a group-specific list but sharing similar structural or linguistic characteristics may also be associated with that group.

Dense text embeddings provide a natural way to formalize these intuitions mathematically and at scale. They represent names as vectors in a high-dimensional space, placing structurally, linguistically, or otherwise similar names closer together based on patterns learned from the corpora used to train the embedding model (Ye et al., 2017; Ye and Skiena, 2019; Wang et al., 2024). As a result, the embeddings derived from a group-specific name list encode information about the similarity of names both on and off the list, allowing us to identify names that are likely to belong to the same group even if they do not explicitly appear in the original list.

Suppose that for each name S , we obtain a d_E -dimensional text embedding vector $E = E(S)$ that captures its linguistic and structural features. Dasanaïke and Imai (2026) demonstrates the value of an embedding-based BISG approach (eBISG), which uses Census name tables to learn $\mathbb{P}(R = r \mid E, G)$ and generalizes the resulting model to rare names whose frequencies are unavailable.

In our setting, however, no such name–group frequency data are available. Instead, we formulate a new proxy prediction task. For each group-specific name list, represented by the indicator L_r , we model the probability that a name with embedding E belongs to the list. Although list membership given name is deterministic (i.e., $L_r \in \{0, 1\}$), the conditional probability of list membership given its embedding, $\mathbb{P}(L_r = 1 \mid E, G)$, is generally nontrivial. We therefore estimate this mapping from embeddings to the probability of list membership using standard supervised machine learning methods.

Unfortunately, this proxy prediction task does not directly recover the target probabilities $\mathbb{P}(R = r \mid E, G)$. Although $\mathbb{P}(L_r = 1 \mid E, G)$ is generally positively associated with $\mathbb{P}(R = r \mid E, G)$, the two quantities are not equal. In Appendix A1, we show that this list membership probability is not well calibrated, which is likely to yield biased downstream estimates. To address this issue, we next show how to recover the target probabilities from the proxy predictions.

2.3 Identifying the target probabilities from proxy prediction

Before introducing the ℓ BISG method, we state the assumptions required for identification. The standard identifying assumption underlying BISG is that, conditional on group membership, name and geography are independent (Elliott et al., 2009).

Assumption 2.1 (BISG).

$$S \perp\!\!\!\perp G \mid R.$$

Because the embedding is a deterministic function of the name, i.e., $E = E(S)$, this assumption implies the identifying assumption used by eBISG (Dasanaïke and Imai, 2026), namely that embedding and geography are conditionally independent given group membership.

Assumption 2.2 (eBISG).

$$E \perp\!\!\!\perp G \mid R.$$

Since the embedding is a lower-dimensional representation of the name, Assumption 2.2 is strictly weaker than the classic BISG assumption (Assumption 2.1).

Under Assumption 2.2, Bayes Rule implies

$$\mathbb{P}(R = r \mid E, G) \propto \mathbb{P}(E \mid R = r) \mathbb{P}(R = r \mid G). \quad (1)$$

Although we assume that $\mathbb{P}(R = r \mid G)$ is identified from an external source such as Census data, no analogous data are available to estimate the embedding distribution $\mathbb{P}(E \mid R = r)$.

The proposed ℓ BISG method is based on the observation that under Assumption 2.2, the following equality holds for each r and g ,

$$\mathbb{P}(E = e \mid G = g) = \sum_{r=1}^{d_{\mathcal{R}}} \mathbb{P}(E = e \mid R = r) \mathbb{P}(R = r \mid G = g). \quad (2)$$

Thus, for each value of e , we have a system of $d_{\mathcal{G}}$ linear equations with $d_{\mathcal{R}}$ unknowns. Under the following rank condition, we can uniquely identify $\mathbb{P}(E = e \mid R = r)$ (Kuroki and Pearl, 2014).

Assumption 2.3 (Rank condition). Let M be the $d_{\mathcal{G}} \times d_{\mathcal{R}}$ matrix with $M_{g,r} = \mathbb{P}(R = r \mid G = g)$. Assume M is full column rank, for which a necessary condition is $d_{\mathcal{G}} \geq d_{\mathcal{R}}$.

Concretely, for each e , let ρ_e be the vector of $\mathbb{P}(E = e \mid G = g)$ and let π_e be the vector of $\mathbb{P}(E = e \mid R = r)$. Then, the target probabilities are found by regressing ρ_e on M :

$$\pi_e = (M^{\top} M)^{-1} M^{\top} \rho_e. \quad (3)$$

Thus, the identification underlying the proposed ℓ BISG method leverages the association between geographic group patterns and name prevalence patterns.

2.4 Leveraging name lists for dimension reduction

While the previous result establishes identification, directly estimating the conditional distribution $\mathbb{P}(E = e \mid G = g)$ is challenging because the embedding vector E is high-dimensional. Our practical solution is to replace the high-dimensional embedding E with an informative lower-dimensional representation $B = B(E)$ that satisfies the following conditional independence assumption:

Assumption 2.4 (ℓ BISG).

$$B \perp\!\!\!\perp G \mid R.$$

Intuitively, B is obtained by clustering or otherwise reducing the dimension of E while retaining information relevant for predicting the target group. Because B is a deterministic many-to-one function of E , Assumption 2.4 is strictly weaker than the eBISG assumption (Assumption 2.2), and therefore also weaker than the classic BISG assumption (Assumption 2.1).

Under Assumption 2.4, equation (2) becomes,

$$\mathbb{P}(B = b \mid G = g) = \sum_{r=1}^{d_{\mathcal{R}}} \mathbb{P}(B = b \mid R = r) \mathbb{P}(R = r \mid G = g). \quad (4)$$

This system consists of $d_B \times d_G$ equations with $(d_B - 1) \times d_R$ unknowns, and the rank condition for identification remains unchanged (Assumption 2.3). When B is a low-dimensional categorical variable, the left-hand side, $\mathbb{P}(B = b \mid G = g)$, can be estimated directly from the data. Solving the resulting system identifies $\mathbb{P}(B = b \mid R = r)$ for all b and r . Bayes' rule then yields the desired target posterior probabilities,

$$\mathbb{P}(R = r \mid B = b, G = g) \propto \mathbb{P}(B = b \mid R = r) \mathbb{P}(R = r \mid G = g),$$

where the proportionality constant implies these must be normalized so that the posterior probabilities of group membership sum to one over r .

While any choice of B yields consistent estimates of probabilities of group membership under Assumptions 2.3 and 2.4, the efficiency of the resulting estimator depends on how informative B is about the target variable R . In particular, greater correlation between $\mathbb{P}(B = b \mid G = g)$ and $\mathbb{P}(R = r \mid G = g)$ leads to more efficient estimation.

To construct such a variable B , we leverage the name lists and the proxy list membership prediction task described in Section 2.2. Specifically, let $f_{r,e} = \mathbb{P}(L_r = 1 \mid E = e)$, and define $\mathbf{f}_e = (f_{e,1}, \dots, f_{e,d_R})^\top$, the vector of predicted probabilities that a name with embedding E belongs to each group-specific name list. We then apply the K -means clustering algorithm to the log-odds transformation of $\mathbf{f}_e$, partitioning them into $K \geq d_R$ clusters with similar patterns of list membership probabilities. When the name lists are informative, B retains substantial predictive information about the target variable. In this case, equation (4) yields a system of $d_G \times K$ equations with $d_R \times (K - 1)$ unknowns. The intuition for why using more than d_R clusters can be helpful is that sometimes, different sub-groups of group r may form distinct sub-clusters. For example, we observe this among Asian surnames in the Florida data example (Appendix A2).

We formally present the ℓ BISG method in Algorithm 1.

2.5 Choosing the number of clusters

We choose the number of clusters K to make B as informative as possible about R without overfitting: too few clusters leave B unable to isolate less common groups, whereas too many clusters introduce sparsity, such that $\mathbb{P}(B \mid R)$ cannot be estimated reliably. Unfortunately, because the clustering is not supervised by R , we cannot select K by held-out accuracy in the usual way. Therefore, we instead select K based on the stability of the resulting posterior probabilities across geographical units since under Assumption 2.4 $\mathbb{P}(B \mid R)$ should not vary as a function of G (see e.g., Lange et al., 2004; Tibshirani and Walther, 2005; von Luxburg, 2010, for similar stability-based approaches).

Specifically, we use group J -fold cross-validation: we partition the geographic units into $J \geq 3$ folds and estimate $\mathbb{P}(B \mid R)$ separately within each fold. Each observation is then scored by the fits from the $J - 1$ folds that exclude its unit, yielding $J - 1$ independent out-of-sample posterior probability vectors. For each observation, we compute the mean of these posterior probabilities, which are sharp (i.e., close to 0 or 1) only when the $J - 1$ held-out fits are both individually confident and mutually consistent. Lastly, we choose the value of K that minimizes the average entropy of each individual's mean posterior probability vectors. Algorithm 2 summarizes the procedure.

Algorithm 1 ℓ BISG

Require: Target dataset of N observations $\{S_i, G_i\}_{i=1}^N$,
Geographic estimates $\hat{\mathbb{P}}(R = r \mid G = g)$,
Name lists $L_r(S)$ for every group $r = 1, \dots, d_{\mathcal{R}}$,
Pre-trained embedding model $E(S)$,
Number of clusters K , chosen by Algorithm 2.

- 1: Obtain name embeddings for each observation i :
 $E_i = E(S_i)$
 - 2: Fit a separate predictive model for membership in each list r :
 $\hat{f}_{e,r} = \hat{\mathbb{P}}(L_r = 1 \mid E = e)$
 - 3: Compute the predicted probabilities for each observation i :
 $\hat{f}_{E_i} = (\hat{f}_{E_i,1}, \dots, \hat{f}_{E_i,d_{\mathcal{R}}})^\top$
 - 4: Construct K clusters through K -means on the log-odds scores z_i , and assign each observation i to a cluster:
 $z_{ir} = \log\{\tilde{f}_{ir} / (1 - \tilde{f}_{ir})\}, \quad \tilde{f}_{ir} = \min\{\max_r\{\hat{f}_{E_i,r}, \epsilon\}, 1 - \epsilon\}, \quad \epsilon = 10^{-6}$
 $B_i = B(E_i) \in \{1, \dots, K\}$
 - 5: Calculate empirical cluster proportions within each geography:
 $\hat{\mathbb{P}}(B = b \mid G = g)$
 - 6: Set vector $\hat{\rho}_b$ and matrix $\hat{M}$ with:
 $\hat{\rho}_{b,g} = \hat{\mathbb{P}}(B = b \mid G = g)$
 $\hat{M}_{g,r} = \hat{\mathbb{P}}(R = r \mid G = g)$
 - 7: Obtain probability estimates $\hat{\mathbb{P}}(B = b \mid R = r)$ as:
 $\hat{\pi}_b = (\hat{M}^\top \hat{M})^{-1} \hat{M}^\top \hat{\rho}_b$
 - 8: Compute the target posterior probabilities:
 $\hat{\mathbb{P}}(R = r \mid B = b, G = g) \propto \hat{\mathbb{P}}(B = b \mid R = r) \hat{\mathbb{P}}(R = r \mid G = g)$
-

This objective $Q(K)$ defined in equation (5) can be written as the sum of the average held-out entropy and the average cross-fold stability, i.e.,

$$Q(K) = \underbrace{\frac{1}{n} \sum_{i=1}^n \frac{1}{J-1} \sum_{j: i \notin \mathcal{Z}_j} H(\mathbb{P}^{(j)}(R \mid B_i, G_i))}_{\text{average held-out entropy}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \text{JSD}(\mathbb{P}^{(1)}(R \mid B_i, G_i), \dots, \mathbb{P}^{(J)}(R \mid B_i, G_i))}_{\text{average cross-fold divergence}},$$

where JSD is the generalized Jensen-Shannon divergence. The first term is smaller when the clusters are more informative. The second term is larger once the clusters are too sparse for the independent fits to agree, so minimizing $Q(K)$ balances informativeness against instability, thereby jointly rewarding the sharpness and stability of posterior predictions. Therefore, unlike a simple entropy criterion, this procedure need not monotonically favor increasingly large numbers of clusters. Appendix A3 shows that on the Florida voter file, in-sample and single-fit held-out entropy both favor more clusters than $Q(K)$.

Algorithm 2 Choosing the number of clusters K

- Require:** List-score vectors $\hat{f}_{E_i}$ and geographies G_i from Step 3 of Algorithm 1,
Geographic prior $\hat{\mathbb{P}}(R = r \mid G = g)$,
Candidate set $\mathcal{K}$ for the number of clusters,
Number of folds $J \geq 3$
- 1: Partition the geographic units into J folds $\mathcal{Z}_1, \dots, \mathcal{Z}_J$ of equal population
 - 2: **for** $K \in \mathcal{K}$ **do**
 - 3: Fit K -means on the log-odds $\{z_i\}_{i=1}^N$ of Algorithm 1 and assign each observation to one of the resulting clusters B_i
 - 4: **for** $j = 1, \dots, J$ **do**
 - 5: Obtain $\hat{\mathbb{P}}^{(j)}(B = b \mid R = r)$ by steps 5–7 of Algorithm 1, using only units in $\mathcal{Z}_j$
 - 6: For every observation i that does not belong to $\mathcal{Z}_j$, compute the out-of-sample posterior $\hat{\mathbb{P}}^{(j)}(R = r \mid B_i, G_i)$, using step 8 of Algorithm 1
 - 7: **end for**
 - 8: For each observation i , compute the average posterior over the $J - 1$ folds:

$$\bar{p}_i := \frac{1}{J-1} \sum_{j: i \notin \mathcal{Z}_j} \hat{\mathbb{P}}^{(j)}(R \mid B_i, G_i)$$

- 9: Compute the average entropy of the averaged out-of-sample posterior

$$Q(K) := \frac{1}{N} \sum_{i=1}^N H(\bar{p}_i), \tag{5}$$

where H is the Shannon entropy

- 10: **end for**
 - 11: **return** $K^* = \arg \min_{K \in \mathcal{K}} Q(K)$
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2.6 Incorporating multiple name lists for each group

The ℓ BISG framework naturally accommodates multiple name lists (e.g., first-name and surname lists) for each group. Let E_F and E_S denote the first-name and surname embeddings, respectively. Applying ℓ BISG separately to each field yields the calibrated probabilities $\mathbb{P}(R \mid B_F, G)$ and $\mathbb{P}(R \mid B_S, G)$, where B_F and B_S denote the first-name and surname clusters.

The standard assumption underlying the incorporation of first names into BISG (known as BIFSG) is that first and surnames are conditionally independent given race and geography (Voicu, 2018). For ℓ BISG, we instead require only the following weaker assumption:

Assumption 2.5 (ℓ BIFSG).

$$B_F \perp\!\!\!\perp B_S \mid R, G.$$

Because the clusters are deterministic functions of the embeddings, this assumption is strictly weaker than conditional independence of the names themselves. Under the assump-

tion,

$$\mathbb{P}(R \mid B_F, B_S, G) \propto \frac{\mathbb{P}(R \mid B_F, G) \mathbb{P}(R \mid B_S, G)}{\mathbb{P}(R \mid G)},$$

where the proportionality constant normalizes the probabilities to sum to one across all groups.

2.7 Discussion

We emphasize that throughout, ℓ BISG imposes no assumptions on the quality of the list or the ranking. Indeed, when compared to the standard BISG method, the only additional assumption required for ℓ BISG is Assumption 2.3, which is typically satisfied if there is some geographic variation in group prevalence. The caveat is that only good lists are able to improve upon the geographic baseline $\mathbb{P}(R = 1 \mid G)$.

In the ℓ BISG algorithm, one could even cluster directly on embeddings E without using name lists. However, such an unsupervised approach relies upon the embeddings being primarily an encoding of group membership so that they are spatially separated in $\mathbb{R}^{d_\varepsilon}$ into clusters that correspond to the group of interest. By comparison, our semi-supervised approach first projects the embeddings to a lower-dimensional metric specifically constructed for the group prediction task. Neither approach produces clusters with known labels, but our clusters leverage name lists to construct more informative clusters.

Empirically, we find that the lists carry important information. For example, in the Florida voter file application, applying K-means clustering to the raw embeddings results in reduced classificatory discrimination for every race, with the highest cost where the lists matter most. Specifically, the average precision for the rare Asian group falls from 0.35 when clustering on list-based scores to 0.10 for raw embedding-based scores (see Appendix A4).

The clustering step of ℓ BISG can also be viewed as using the embeddings to update and extend our original lists. For example, if Hernandez originally appeared on the Hispanic list and Fernandez did not, their closeness in embedding space hopefully will mean that Fernandez is also included in the Hispanic cluster. In this way, we generalize initial, possibly quite small, lists to the entire embedding space.

Finally, researchers often have lists for only a few groups of interest rather than a full partition of the population. Because the subgroup probabilities sum to one, it is sufficient to estimate them for the $d_{\mathcal{R}} - 1$ named groups, leaving a residual class as the complement. That said, forming an explicit residual list for that class, from names distinctive of the population but of no named group, can be worthwhile. In Section 3, we show that the provision of such a name list becomes necessary once the subgroup geography must be recovered rather than assumed known.

Another advantage of ℓ BISG is that it naturally accommodates overlapping name lists, in which the same name appears on multiple lists. Specifically, each probability $\mathbb{P}(L_r \mid E = e)$ is estimated as a separate prediction task, treating each name separately for each target group. Consequently, a name appearing on multiple lists simply receives a high predicted probability for each corresponding task, and this may be reflected in the clustering, which is performed on a function of the full vector of predicted list membership probabilities, $\hat{f}_e$. Although overlap makes the target groups less distinguishable, it does not otherwise affect the validity of the method.

3 Recovering Subgroup Probabilities

We next consider a common but challenging setting in which geographic prevalence data $\mathbb{P}(R \mid G)$ are not available for groups of interest, though such information exists for coarser groups. For example, the U.S. Census provides geographic prevalence for the broad Asian category but not for finer subgroups such as Chinese, Korean, Japanese, or South Asian. This creates a barrier to directly applying the ℓ BISG method of Section 2 with subgroup name lists.

In this section, we show how to combine subgroup-specific name lists with coarse-group geographic prevalence rates to recover fine-grained subgroup probabilities through ℓ BISG. Unlike the preceding application of ℓ BISG, this procedure requires perfect separation across subgroup name lists: no member of one subgroup may have a name on another subgroup’s list. A necessary, but not sufficient, condition for satisfying this perfect separation assumption is that the name lists do not overlap. Under this additional assumption, we can first recover subgroup geographic probabilities and then apply the ℓ BISG method described in Section 2.

The method introduced in this section is therefore applicable only to subgroups whose names are distinct from one another. In Appendix A5, we develop a sensitivity analysis tool for relaxing the perfect-separation assumption and assessing how this affects the empirical results. However, we recommend these only for cases where the expected violations are fairly small and homogeneous across groups.

3.1 The problem setup

Let $A \in \mathcal{A}$ denote a finer subgroup variable that partitions the values of race R , so that each subgroup value a belongs to a unique coarse group $r = r(a)$. Let $d_{\mathcal{A}}(r)$ denote the number of subgroups within coarse group r and $d_{\mathcal{A}} = \sum_r d_{\mathcal{A}}(r)$ be the total number of subgroups. The challenge is that geographic prevalence rates $\mathbb{P}(A = a \mid G)$ for the subgroup variable A are unavailable. Instead, we only have the geographic baseline for the coarse group $\mathbb{P}(R = r \mid G)$.

We assume that a name list, represented by list indicator L_a , is available for each subgroup of interest a . Indeed, unlike before, we require all $d_{\mathcal{A}}(r)$ subgroups to have their own lists (i.e., $d_{\mathcal{A}}(r) - 1$ lists is not sufficient). In other words, this set of sub-groups must exhaustively span each coarser group. This is to ensure that they sum to the coarse group baseline, i.e., $\mathbb{P}(R = r \mid G) = \sum_{a=1}^{d_{\mathcal{A}}(r)} \mathbb{P}(A = a \mid G)$. If, in practice, there are not $d_{\mathcal{A}}(r)$ lists, but researchers do have access to an overall list for the coarse group, then we add a residual class whose list is the coarse group’s name list minus every name claimed by any of the subgroup lists. The residual name list then enters the embedding step as an ordinary list, contributing its own score to $\hat{f}_e$.

In the following sections, we propose a solution to the problem of identifying $\mathbb{P}(A = a \mid G)$ that relies on two additional assumptions: one about the construction of the name lists and one which relates to both list quality and the underlying degree of separation of the groups by their names.

3.2 List construction must not depend on geography

The first additional assumption is that conditional on finer-group membership, list membership is independent of geography.

Assumption 3.1 (List construction).

$$L_{a'} \perp\!\!\!\perp G \mid A = a \quad \text{for all } a, a' \text{ such that } r(a) = r(a') = r. \quad (6)$$

This assumption is implied by the standard BISG assumption (Assumption 2.1) provided that list membership is determined solely by name and selected from a full pool of names from across geographic units.

The assumption may be violated, however, if only the weaker eBISG or ℓ BISG assumptions hold (Assumption 2.2 and 2.4, which concern functions calculated from names) so that at the resolution of the original raw names, some geographic specificity also affects the way the list is compiled. For example, suppose a list is compiled from a geographically unrepresentative source, such as the membership rolls of a regional organization. In that case, even among individuals in the same finer subgroup, list membership may remain associated with geography (see Section 4.1 for a discussion of list construction and evaluation). One can address this problem by conditioning on a coarser geographic unit, thereby requiring the assumption to hold only within each such unit.

For the sub-group recovery problem, assumption 3.1 implies that $\mathbb{P}(L_{a'} = 1 \mid A = a, G)$ does not vary with G . Hence the observable list prevalence by geography $\mathbb{P}(L_{a'} = 1 \mid G)$ satisfies, by the law of total probability,

$$\mathbb{P}(L_{a'} = 1 \mid G) = \sum_{r=1}^{d_{\mathcal{R}}} \sum_{a:r(a)=r} \mathbb{P}(L_{a'} = 1 \mid A = a) \mathbb{P}(A = a \mid G), \quad (7)$$

where the list indicator plays the role that embeddings played in equation (2). If the within-subgroup list membership rates $\mathbb{P}(L_{a'} = 1 \mid A = a)$ were known, this would again form a system of linear equations that is identified under rank conditions.

Unfortunately, equation (7) cannot recover both the within-subgroup list rates and the subgroup geographic prevalence simultaneously. Although the subgroup prevalence is constrained to sum to the known coarse-group prevalence, i.e., for each r ,

$$\mathbb{P}(R = r \mid G) = \sum_{a:r(a)=r} \mathbb{P}(A = a \mid G), \quad (8)$$

these additional constraints are not sufficient to uniquely identify the system.

3.3 Proposed solution given perfect separation across subgroup name lists

The within-group list membership rates $\mathbb{P}(L_{a'} = 1 \mid A = a)$ reflect how well the lists separate the subgroups. When $a \neq a'$, a positive value of one of these cross-group-rates indicate that there are members of one sub-group a whose names appear on the list for a sub-group a' . This can happen for two reasons: (1) poor list quality and (2) inherent lack of group separation by names. For example, if trying to separate English and Irish sub-groups, positive cross-group-rates could be caused by using too-expansive lists not designed for the

purpose of separating these groups. At the same time, historical and linguistic ties between the two groups may also make these groups inherently harder to separate.

To start, we assume that lists perfectly separate the groups: no member of one sub-group has a name on the list of another.

Assumption 3.2 (Exclusive lists.). $\mathbb{P}(L_{a'} = 1 \mid A = a) = 0$ for all $a \neq a'$

Assumption 3.2 allows incomplete coverage (i.e., $\mathbb{P}(L_a = 1 \mid A = a) < 0$). It also remains possible for there to be members of groups a' and a that share the same name, but those names cannot appear on either group's list of distinctive names. Under this assumption, equation (7) becomes simply

$$\mathbb{P}(L_{a'} = 1 \mid G) = \mathbb{P}(L_a = 1 \mid A = a) \mathbb{P}(A = a \mid G), \quad (9)$$

and combined with equation (8), we show in Appendix A5 that the geographic rates have the following unique solution for each subgroup r .

$$\mathbb{P}(A = a \mid G = g) = \frac{\mathbb{P}(L_{a'} = 1 \mid G = g)}{((\beta_r^\top \beta_r)^{-1} \beta_r^\top \mathbf{M}_r)_a} \quad (10)$$

where β_r is the $d_G \times d_A(r)$ matrix of list rates $\mathbb{P}(L_a = 1 \mid G)$ in each geography, and where $\mathbf{M}_r$ is the length d_G vector of coarse group prevalence rates $\mathbb{P}(R = r \mid G = g)$. This formula takes the sub-group list prevalence in geography g and adjusts it by a denominator which captures the average relationship between the sub-group list prevalence and coarse group prevalence across geographies.

We acknowledge that Assumption 3.2 may not hold exactly in practice. In Appendix A5, we provide sensitivity analysis tools for probing how results change under small violations of assumption 3.2. Specifically, we show how to adjust the denominator of equation (10) if the cross-probabilities instead take a small value $\lambda > 0$ either uniformly overall or uniformly within each coarse group. However, this sensitivity analysis still does not capture what would happen for larger heterogeneous violations of Assumption 3.2.

We caution that performance can be poor under heterogeneous non-separation in name lists between sub-groups. To make progress in such cases, researchers will need to gather geographic information about subgroups and apply the ℓ BISG method described in Section 2. Alternatively, they might considering pooling strongly overlapping sub-groups (e.g., English and Irish) and focusing on recovery for the remaining more distinct groups.

Ultimately, for well-separated groups, the above method allows us to recover $\mathbb{P}(A = a \mid G)$, which are already an important improvement on $\mathbb{P}(R = r \mid G)$ when subgroups of interest. Once recovered, these can be further improved by inputting them into the overall ℓ BISG method of Section 2.

4 Obtaining and Evaluating Name Lists

In this section, we briefly discuss how to obtain and evaluate group-specific name lists.

4.1 Generating name lists

There are different ways to build name lists. First, lists can be curated from domain knowledge or existing records. For example, they may be constructed from prior surveys, censuses, or genealogical records; membership rolls of group-affiliated organizations; obituaries associated with group-specific funeral homes (Chasalow et al., 2026); curated Wikipedia pages; or the activities of identified users on social media. Curated lists can be highly distinctive, but their coverage depends on the source. Sources tied to a particular place may violate the no-geographic-selection assumption (Assumption 3.1) that the subgroup-geography recovery of Section 3 relies on.

A second approach, which makes ℓ BISG applicable even when no frequency table or curated record exists, is to prompt an LLM to generate surnames typical of a target group, thereby leveraging the world knowledge encoded during pre-training (Dasanaïke, 2026). This approach is broadly applicable across many types of groups and targets. Because the model returns the names it judges to be typical, the resulting lists may be common but also include surnames that are not distinctive.

To improve distinctiveness, one can apply a *within-group exclusivity filter* by removing any surname that appears on more than one group’s list. This is essential for the subgroup recovery method described in Section 3. In addition, one can prompt an LLM to generate surnames that are distinctively associated with the group of interest. The generated list can then be refined by asking a possibly different model to assess the distinctiveness of each name and retaining only those that are judged to be truly distinctive.

4.2 Desirable properties of name lists

What makes a good name list? Ideally, a name list contains names that are both *distinctive* and *common* within the group of interest. Distinctive names have high positive predictive value for the target group (i.e., $\mathbb{P}(R = r \mid S = s)$ is high), whereas common names have high coverage of the target group (i.e., $\mathbb{P}(S = s \mid R = r)$ is high).

To illustrate, we generate a name list for each of the four racial groups in the Florida voter file of Section 5.1 using an LLM, Gemini 3.0 Flash. This corresponds to a realistic setting, in which researchers must create lists on their own without relying on name frequency tables. To ensure diversity within each racial group, we use separate prompts for different national origins. For the White list, for instance, we use three prompts, each requesting surnames of individuals with English and Irish descent; German descent; and Italian, Polish, and other European descent. We split Asian into Chinese; Indian; Vietnamese and Filipino; and Korean and Japanese. Appendix A7 provides the exact prompts.

For each of these prompts, we make six separate model calls at temperature 1.0, and request 300 surnames per call. Six calls therefore produces 5,400 surnames each for the White, Black, and Hispanic groups, while the Asian group yields 7,200 surnames. We rank the unique surnames by the number of calls that returned them and then retain the top 1,000 surnames. The gain in discrimination from longer lists is essentially complete at a few hundred surnames per group (see Figure A9 in Appendix A8.2). This means that a surname ranks higher when the model produces it consistently. Because that score is a count over eighteen calls, or twenty-four for Asian surnames, many surnames are ranked in a tie. Since the top 1,000 threshold often falls inside a tied band, we break those ties by the order in



Figure 1: Florida surnames placed by commonness $P(S | R)$ (x -axis) against distinctiveness $P(R | S)$ (y -axis) for the four race groups. Each point is a surname held by at least twenty-five members of the group, sized by how many carry it; colored points are those included in the list of 1,000 surnames generated by an LLM for that group. Labeled points are the most common surnames in each group. The dashed line marks the distinctiveness floor of 0.3 used to build lists in Section 4.1.

which the model first returned each surname.

Figure 1 plots the surnames held by Floridians of each race along those two dimensions: distinctiveness (y -axis) and commonness (x -axis). Each point represents a surname held by at least 25 individuals, and the colored points indicate the surnames included in the LLM-generated list of 1,000 surnames for that group.

The four racial groups exhibit distinct patterns on the same axes. White surnames span a broad range of both commonness and distinctiveness. Hispanic surnames form a dense band characterized by high distinctiveness and a range of commonness, making them well suited for both identifying and representing the group. Black surnames are generally less distinctive because they are more frequently shared with other groups, in particular white, although some are quite common. Asian surnames resemble Hispanic surnames in being both relatively common and highly distinctive, with some exceptions (e.g., “Lee”). The LLM generally selects surnames that are both distinctive and common. The exception is

Black surnames, for which it prioritizes commonness over distinctiveness (see Section 4.1 for further discussion).

Commonness matters for a second reason. The embedding step extends the information in a list to unlisted names that are close in embedding space. A list whose names provide broader coverage of the group’s name distribution thus yields more accurate probabilities for unlisted names. In Appendix A2, we visually inspect list coverage in embedding space.

4.3 Measure of list quality

Although the relationship between list membership and true group membership cannot be directly measured, under Assumption 3.1 of no geographic selection in the lists, it is possible to recover probabilities $\pi_{r,r'} := \mathbb{P}(L_{r'} = 1 \mid R = r)$, using the same strategy as the one used for recovering $\mathbb{P}(B = b \mid R = r)$. That is, we solve the following linear system of $d_{\mathcal{G}}d_{\mathcal{R}}$ equations with $d_{\mathcal{R}}^2$ unknowns,

$$\mathbb{P}(L_{r'} = 1 \mid G = g) = \sum_{r=1}^{d_{\mathcal{R}}} \pi_{r,r'} \mathbb{P}(R = r \mid G = g). \quad (11)$$

Let $\mathbf{\Pi}$ be the $d_{\mathcal{R}} \times d_{\mathcal{R}}$ matrix of these values with $\mathbf{\Pi}_{r,r'} = \pi_{r,r'}$. One measure of the quality of $\mathbf{\Pi}$ is diagonal dominance:

$$\gamma(\mathbf{\Pi}) = \frac{1}{d_{\mathcal{R}}} \sum_{r=1}^n \left[\pi_{r,r} - \frac{1}{d_{\mathcal{R}} - 1} \sum_{r' \neq r} \pi_{r,r'} \right] = \frac{\text{trace}(\mathbf{\Pi}) - 1}{d_{\mathcal{R}} - 1}. \quad (12)$$

which is maximized at the ideal $\pi_{r,r'} = \mathbf{1}(r = r')$ with value 1 and minimized at a pathological scenario of $\pi_{r,r'} = \mathbf{1}(r \neq r')$ with value -1 . Positive values indicate a tendency for diagonal $\pi_{r,r}$ values to be larger than the off-diagonal values.

In the binary group case where $R = 1$ indicates the group of interest and $R = 0$ is simply the residual group with no list, we can solve for $\pi_1 = \mathbb{P}(L = 1 \mid R = 1)$ and $\pi_0 = \mathbb{P}(L = 1 \mid R = 0)$ and measure quality as $\pi_1 - \pi_0$.

The recovered matrix also separates the two components combined by γ . The diagonal entry $\pi_{r,r}$ is the coverage of list r among members of group r . Applying Bayes’ rule to column r , with the marginal group shares $q_{r'} = \mathbb{P}(R = r')$ obtained from the geographic prior, yields the precision of the list, $\mathbb{P}(R = r \mid L_r = 1) = q_r \pi_{r,r} / \sum_{r'} q_{r'} \pi_{r',r}$. Both quantities can therefore be recovered without labels. In the validation exercises below, we report the estimated and ground-truth coverage and precision.

Although our proposed method neither requires Assumption 3.1 nor presupposes any particular level of list quality, estimating $\mathbf{\Pi}$ can provide a useful way to assess list quality when true labels are unavailable for computing other evaluation metrics. This metric rewards lists that both cover their intended group and remain exclusive to it.

5 Empirical Validation

We empirically validate ℓ BISG in three diverse settings with known ground truth. First, we use voter files from Florida and North Carolina, where self-reported race is available.

We compare the performance of ℓ BISG with that of the standard BISG based on Census name frequency tables. Second, we use the full-count 1900 U.S. Census to predict Chinese and Japanese subgroups within the Asian population, to evaluate the performance of the methodological extension proposed in Section 3. This represents a level of granularity for which no name frequency tables naturally exist; Appendix A6 applies the same extension to white country-of-origin subgroups, where the lists are not exclusive to the coarse group. Third, we move beyond the United States and consider the prediction of religious sect in Lebanon, illustrating the applicability of ℓ BISG outside the conventional race-based BISG setting. In each application, we evaluate the calibration of the predicted probabilities, their discriminatory power, and the bias and RMSE of downstream disparity estimates.

5.1 Comparison with standard BISG using US voter files

Setup. We obtain Florida and North Carolina voter files from L2 Inc., a leading national non-partisan firm and the oldest organization in the United States that supplies voter data and related technology to candidates, political parties, pollsters, and consultants for use in campaigns. These voter files contain the self-reported race of each registrant alongside their geocoded residential address, turnout in recent elections, and political party registration.

We collapse the race field into five categories, White, Black, Hispanic, Asian, and other, the last of which includes American Indian and Alaska Native. We use the Florida file of March 7, 2026 and the North Carolina file of April 11, 2026. We exclude registrants whose race is not recorded in the voter file and those without geocoded coordinates, comprising 494,443 and 20,323 of Florida’s 15,017,471 voters and 136,730 and 2,652 of North Carolina’s 8,609,276, respectively. Lastly, we draw a random sample of one million registrants from the remainder in each state.

We apply ℓ BISG and standard BISG with the published Census tract-level race shares as the geographic priors. In the downstream analysis, we use these tract-level predictions to estimate the Democratic share of each racial group aggregated at the county level, based on party registration. The county-level comparison ensures that race-by-county cells are large enough for the evaluation metrics to be stable.

To estimate the individual race probabilities using ℓ BISG, we draw on two sets of name lists in turn. The first name list is based on the 2020 Census surname table, which reports the racial composition of every surname occurring at least 100 times in the Census. Because this list is directly derived from the Census data, we can evaluate the consequences of losing frequency information on the quality of individual race probability estimates and downstream analysis by comparing the resulting estimates with those based on the standard BISG with the surname table.

As described in Section 4.2, a useful list has names that are sufficiently common and yet distinctive, with clear group membership. For each group r , we construct the corresponding Census list by filtering to surnames that are at least 30% likely to belong to that group, i.e., $\mathbb{P}(R = r \mid S = s) \geq 0.30$.¹ We then rank each surname by how many group members have that surname, beyond the number expected in a random set of the same sample size, i.e.,

¹We set the floor at 0.30 because it is a smallest value at which the four lists are close to disjoint. For example, 7.0% of Florida registrants have a surname belonging to more than one list at a floor of 0.20, whereas this proportion is only 0.04% at a floor of 0.30. Raising the floor further reduces coverage among Black registrants while leaving the other three lists almost unchanged (see Table A4 in Appendix A8.2).

Group	List	In Census	Coverage		Precision	
			Ground truth	Estimate	Ground truth	Estimate
White	Census	100%	0.28	0.30	0.79	0.74
	LLM	99%	0.22	0.24	0.73	0.70
Black	Census	100%	0.33	0.37	0.46	0.46
	LLM	100%	0.50	0.55	0.30	0.29
Hispanic	Census	100%	0.50	0.53	0.82	0.97
	LLM	99%	0.49	0.53	0.81	0.97
Asian	Census	100%	0.42	0.41	0.64	0.73
	LLM	96%	0.25	0.21	0.17	0.18

Table 1: Comparison of LLM-based and Census-based surname lists on the Florida registrants. The column labeled as “In Census” shows the share of list names that appear in the 2020 Census surname table at all. “Coverage” is the share of the group’s registrants with a surname on the list, and “Precision” the share of the registrants with a list surname who belong to the group, each computed from the self-reported race (ground truth) and estimated from the tract system with the tract-level race shares as the geographic prior and no individual labels (Section 4.3).

$\mathbb{P}(S = s \mid R = r) - \mathbb{P}(S = s)$. Lastly, we take the 1,000 highest-ranked surnames for each group. Because we construct each list independently across groups, a surname may qualify for more than one list. However, this is rare, as only seven surnames appear on two lists after filtering to the top 1,000.

We obtain the second set of name lists from an LLM, Gemini 3.0 Flash, as described in Section 4.2. For ℓ BIFSG, we construct first-name lists in the same two ways, from the 2020 Census first-name table by the rule above and from the LLM with the same prompts. Finally, we embed surnames using the multilingual E5-large model (Wang et al., 2024) and estimate the predicted probability of being on each list $f_{e,r}$ separately using a multilayer perceptron.² Each model is scored on five out-of-fold splits, so that no surname contributes to the score that it receives. We then apply the ℓ BISG method via Algorithm 1, selecting the number of clusters with Algorithm 2, which returns $K = 150$ for the LLM lists and $K = 75$ for the Census lists in Florida, with cross-fitting by individual voters. We fit standard BISG using the same 2020 Census surname table and geographic priors.

Quality of name lists. Before we examine the performance of ℓ BISG, we investigate the quality of LLM name lists relative to that of the Census-based lists, which can be thought of as a gold standard. As shown in the column labeled “In Census” in Table 1, between 96 and 100% of the LLM surnames generated for each group appear in the Census table, indicating that the model reliably produces plausible names. We note that the Census released the 2020 surname table in April 2026 and that Gemini 3.0 Flash was trained only on data through January 2025, so the validation data is not included in the training data. The 2000 and 2010 surname tables have been public for much longer, and so the model may be aware of them.

The two name list constructions diverge in coverage and precision, in particular for the Black and Asian lists. For Black registrants, the LLM covers 0.50 of the group against 0.33

²The model has two hidden layers of 256 and 128 units, ReLU activations, and dropout of 0.3. We use PyTorch for eight epochs with Adam at a learning rate of 10^{-3} and weight decay 10^{-5} under binary cross-entropy.

for the Census list. It has a precision of 0.30 against 0.46, indicating that the model returns more common but less distinctive surnames when compared to the Census lists. The LLM is worse on both metrics for Asian registrants, 0.25 against 0.42 with respect to coverage and 0.17 against 0.64 for precision. This is in part because the model over-produces unique Korean and Chinese surnames and under-produces South Asian and Filipino ones. For White and Hispanic registrants, the two lists are close on both measures.

In addition, the LLM lists overlap across racial groups much more often than the lists based on the Census surname data: 15.5% of Florida registrants have a surname belonging to more than one LLM list, compared to only 0.04% for the Census-based lists. This makes sense because our Census-based name list is intentionally constructed to increase distinctiveness.

We also report the measure of list quality proposed in Section 4.3, based on the diagonal dominance $\gamma(\Pi)$ of the matrix whose (r, r') entry $\pi_{r,r'}$ is the rate at which members of group r carry a name on list r' (Figure A10). Estimated from the tract-level list hit rates and the published tract race shares alone, γ is 0.36 for the Census lists and 0.32 for the LLM lists. This suggests that the former is of slightly higher quality than the latter. These estimates are substantively similar to their ground truths based on self-reported race, which are 0.34 and 0.29 for the Census and LLM lists, respectively, indicating that the proposed quality measure is reliable.

The estimated Π matrix also recovers coverage and precision well. Table 1 reports both quantities as estimated from the tract hit rates and race shares without individual race labels, next to their ground truths. Estimated coverage is within 0.05 of its ground truth and estimated precision within 0.16 for every list. The largest differences are observed for the Hispanic lists and the Census Asian list, where the estimates overstate precision. The Census lists are more precise than the LLM lists for every group, both in the ground truth and the estimates.

Results. We compare the empirical performance of ℓ BISG with LLM lists and standard BISG for Florida and North Carolina. As shown in Figure 2, ℓ BISG (red solid line) nearly matches the precision and recall of standard BISG (blue dashed line) for Whites and Hispanics in both states, without requiring a detailed frequency table. For Black and Asian voters, standard BISG outperforms ℓ BISG, by more for Asian voters. This reflects the lower precision of the LLM name lists for these two groups (Table 1).

Figure 3 shows that the predicted individual race probabilities based on ℓ BISG are reasonably well calibrated, though less so than those based on standard BISG. For Black voters, both methods are nearly perfectly calibrated. For White voters, ℓ BISG probabilities tend to underpredict this racial group, and for Hispanic voters they tend to overpredict it, in part because the published tract shares are those of the tract population rather than of registered voters. Lastly, for Asian voters, both methods overpredict at high probabilities, and ℓ BISG probabilities are better calibrated even though their discrimination power is low as shown in Figure 2.

We also evaluate the validity of downstream analysis using the individual race probabilities obtained from ℓ BISG. We estimate the Democratic share of registrants in each racial group with BIRDIE (see McCartan et al., 2025, for details). BIRDIE takes the ℓ BISG probabilities as its first stage and uses the surname as an instrumental variable for race,

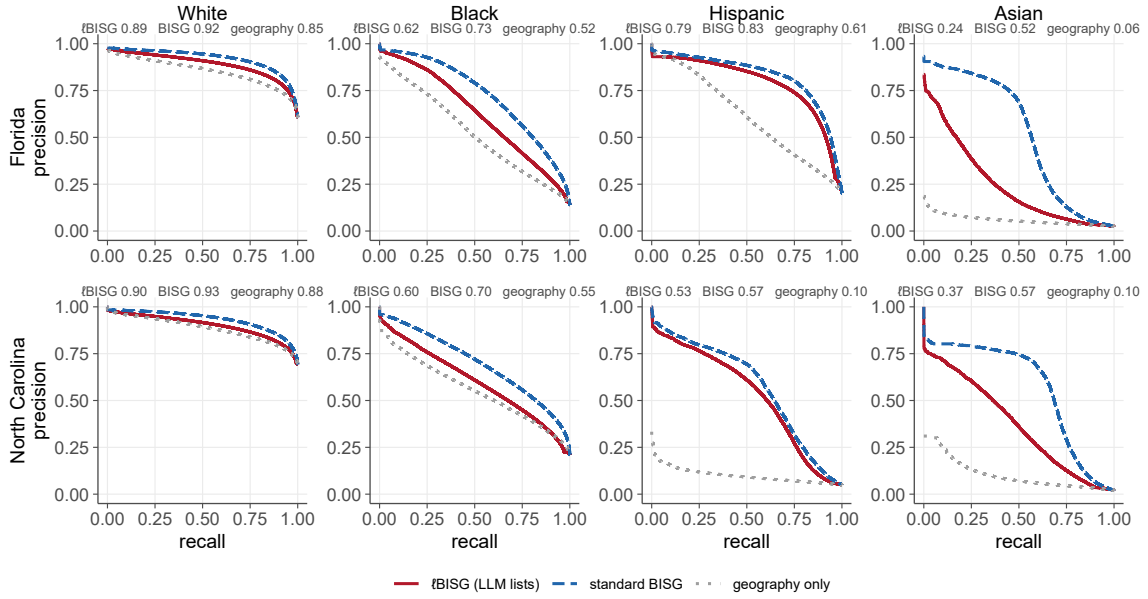


Figure 2: Precision and recall for ℓ BISG (LLM lists, red, solid), standard BISG (blue, dashed) and geography alone (grey, dotted), by state (rows) and race (columns). Average precision is reported above each panel.

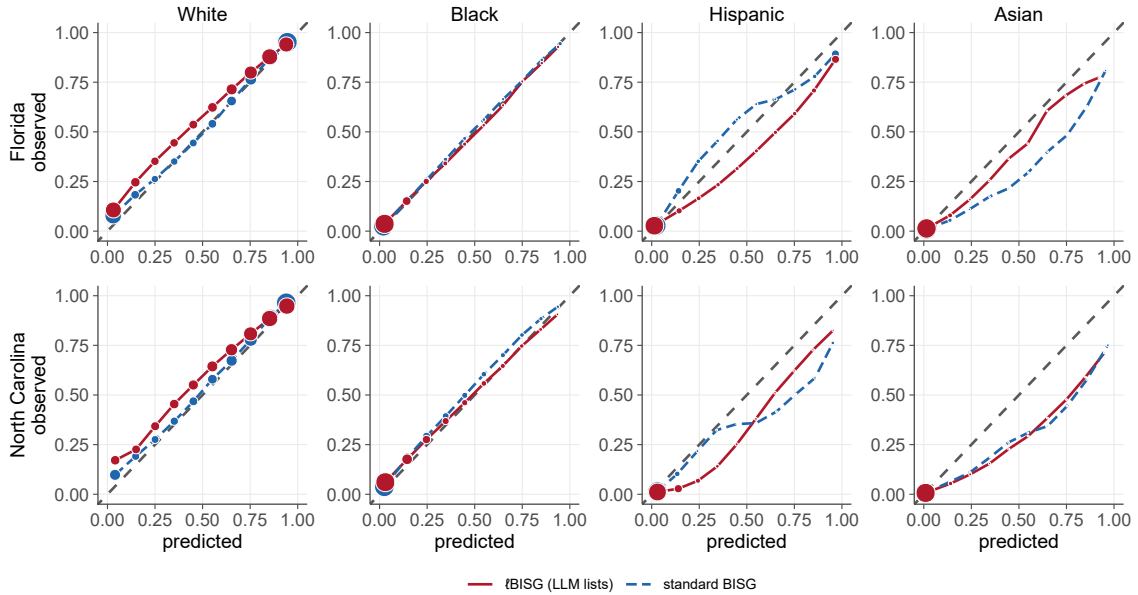


Figure 3: Calibration of estimated individual race probabilities based on ℓ BISG (LLM lists, red, solid) and standard BISG (blue, dashed) for Florida (top row) and North Carolina (bottom row). The dashed diagonal marks perfect agreement between estimated probabilities and observed proportions, and the area of each circle is proportional to the number of registrants in the bin.

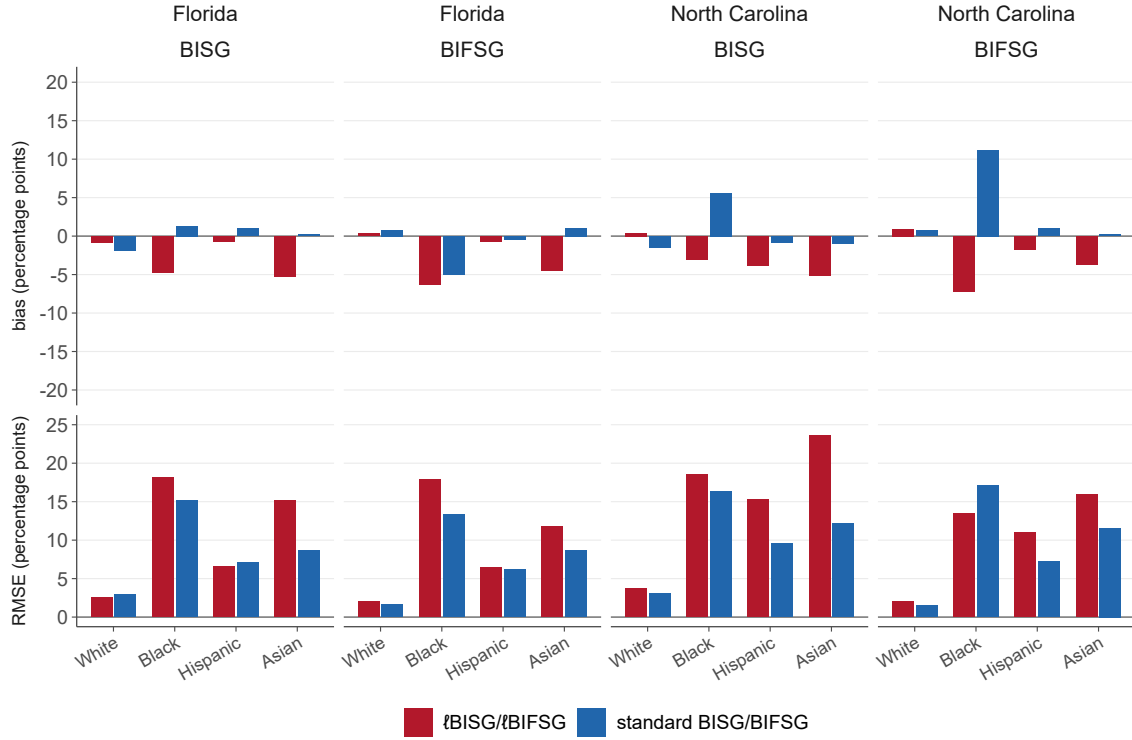


Figure 4: County-level race-specific estimates of the Democratic share of registrants. Rows give signed bias (top) and root-mean-square error (bottom), both in percentage points; the left two columns are Florida and the right two North Carolina, each pair showing the surname-only estimators (BISG) and then those that add first names (BIFSG). Red bars are the ℓ BISG/ ℓ BIFSG (LLM lists) estimators, whereas blue bars are the estimates based on the standard BISG/BIFSG.

assuming that the surname is independent of the outcome conditional on race and geography. This corrects the attenuation bias of the standard downstream estimation approaches that assume conditional independence between race and outcome conditional on name and geography.³ We fit the saturated specification of BIRD_i at the county level, which estimates the Democratic share separately for every racial group within each county with no pooling across counties, since the race-by-county cells hold a sufficient number of registrants alone.

Figure 4 compares the estimated Democratic share of each racial group at the county level against the truth. The top panel shows the bias, where the magnitude of bias is smaller for ℓ BISG (red bars) for White registrants in both states. For Blacks, ℓ BISG has a smaller bias than the standard BISG (blue bars) in North Carolina though the former has a larger bias in Florida than the latter. For Hispanics, the bias is small for both methods in Florida, while ℓ BISG has the larger bias in North Carolina. Asian registrants are the group for which ℓ BISG has a noticeably greater bias than the standard BISG in both states, reflecting the low list quality identified above.

The bottom panel shows the root mean squared error (RMSE). ℓ BISG is within one percentage point (ppt) of standard BISG for White and Hispanic registrants in Florida, and

³This assumption is especially difficult to justify for Black and Whites who have similar surnames.

within one ppt for White registrants in North Carolina. ℓ BISG falls behind standard BISG by 3.1 ppts for Black registrants in Florida and 2.3 ppts in North Carolina, by 5.7 ppts for Hispanic registrants in North Carolina, and by the most for the rarer Asian group, whose name list is weaker, by 6.5 ppts in Florida and 11.4 ppts in North Carolina.

Adding first names (BIFSG) lowers county-level error in every case. In terms of RMSE, ℓ BIFSG improves on ℓ BISG for every group in North Carolina, by 7.7 ppts for Asian registrants and 5.1 ppts for Black registrants, and for every group in Florida, by between 0.2 and 3.3 ppts. For bias, the addition of first names improves the Asian estimate most, from -5.2 to -4.5 ppts in Florida and from -5.1 to -3.8 ppts in North Carolina. In contrast, the Black estimates worsen in both states, from -4.7 to -6.3 ppts in Florida and from -3.0 to -7.2 ppts in North Carolina, suggesting that the reduction in RMSE for that group is due to a decrease in variance. Standard BIFSG likewise improves upon standard BISG in RMSE for six of the eight state-group pairs, the exceptions being Black registrants in North Carolina and Asian registrants in Florida.

We also repeat the downstream analysis using the Census lists (see Figure A11 in the Appendix). The results are similar to those based on the LLM lists shown here, though there are some important differences. For White and Hispanic registrants, the two list constructions fall within 0.9 ppts of one another in county RMSE in Florida and within 2.3 ppts in North Carolina. For Black registrants, the two are within 0.9 ppts in Florida and 0.1 ppts in North Carolina. The difference is greatest for Asian registrants, the group whose LLM list differs most from the Census list: the LLM list produces the more accurate estimate in Florida, by 1.7 ppts, and the Census list in North Carolina, by 4.4 ppts, though both fall well behind standard BISG. The Census lists also yield smaller absolute bias for five of the eight state-group pairs.

Overall, we find that ℓ BISG is comparable to standard BISG in accuracy, calibration, and downstream analysis validity for White and Hispanic registrants, without requiring a frequency table, while first names improve performance further through ℓ BIFSG. The results depend on measurable group-specific list quality. With respect to the LLM lists, the Black list has low precision and the Asian list is weak in both coverage and precision, and for these groups, the downstream analysis based on ℓ BISG is less accurate than that based on standard BISG.

5.2 Predicting Asian subgroups using US historical census

We next turn to the US historical census, which provides population-scale, individual-level records decennially from 1850 to 1950 (Nelson et al., 2025). Records include name and race, and the 1900 census enumerates Chinese and Japanese as separate races. We validate ℓ BISG estimates against the enumerated race of each person. We assume the setting of Section 3, where the target groups are subgroups of a broader group whose geography is known, while the subgroup geography is not. Standard BISG is not applicable here, because neither surname frequency table nor geographic prevalence information exists for these groups.

Setup. We use the de-anonymized (restricted access) 1900 Federal Census. By 1900, 126,000 Chinese and 94,000 Japanese Americans were enumerated living in the US, concentrated in the West and in Hawaii. No other Asian group numbered more than 1,000. We therefore

use the Chinese and Japanese populations as the two target subgroups, and the Asian population as the coarse group.

The published tables of the 1900 census report the size of the Asian population in every county but do not decompose the population by country of origin. We therefore know the county geography of the coarse group but not of the subgroups, which is the setting described in Section 3. We apply ℓ BISG with the county-level geographic prior for the coarse group and LLM name lists. We then conduct a downstream analysis to estimate each subgroup’s socioeconomic standing at the same county level.

The additional assumption for the current setting, in which the geographic prior is available only for the coarse group, is that a person has a surname on a subgroup’s list only if they belong to that subgroup (see Assumption A5.1). Unlike the national-origin application of Appendix A6, this assumption is credible here: Chinese and Japanese surnames are rarely held by people of other races.

We consider two settings. First, we treat the county geography of the two subgroups as known and construct the geographic prior from the census records and apply the ℓ BISG method. We refer to this procedure as “oracle ℓ BISG,” which serves as a benchmark. Under the second and more realistic setting, we treat only the coarse geography as known (the Asian population of each county). We recover the subgroup shares in each county from the list hit rates using the method described in Section 3 (assuming $\lambda = 0$).

We construct a list for each subgroup by prompting an LLM. As in Section 5.1, we also use an LLM, Gemini 3.0 Flash, to assist in writing these prompts. For each subgroup, we ask the model to propose three short descriptions that identify subgroups within that population, then substitute these descriptions into the same skeleton, which asks for surnames “spelled as they were written in American census and immigration records of that period.” For example, for the Chinese population, the descriptions are “Cantonese families from the Pearl River Delta settling in California and the American West,” “Taishanese and Sze Yup laborers migrating to the sugar plantations of the Hawaiian Islands,” and “Fujianese and Hokkien merchants and sailors establishing communities in Atlantic port cities like New York.” Appendix A7 lists the descriptions generated for every group.

For each description, we make six separate model calls at temperature 1.0, requesting 300 surnames per call, rank the unique surnames by the number of calls that returned them, and retain the top 1,000, breaking ties by the order in which the model first returned each surname. To make Assumption A5.1 as credible as possible, we then apply the exclusivity filter across the two lists, dropping the three surnames that appear on both, and drop every surname that also appears on any of the white national-origin lists of Appendix A6. After filtering, the Chinese list holds 933 surnames and the Japanese list 991. Even so, 66% of the people carrying a surname on the Chinese list and 21% of those carrying a surname on the Japanese list are not Asian, mostly white Americans named Lee, Young, or Chin.

The remaining steps follow Algorithm 1. We embed surnames using the multilingual E5-large model and estimate $\hat{f}$ in the same way as done in Section 5.1. Algorithm 2 returns $K = 3$ under both the oracle and the recovered prior, with cross-fitting by individual records.

Quality of name lists. Table 2 reports coverage and precision for each list computed from the enumerated race. The Chinese list covers 0.61 of Chinese records with a precision of 0.34,

Subgroup	Records	Coverage		Precision
		Ground truth	Estimate	Ground truth
Chinese	114,095	0.61	0.64	0.34
Japanese	87,273	0.26	0.24	0.79

Table 2: Quality of the two LLM subgroup lists on the Asian population of the full-count 1900 census. “Records” counts the people enumerated as Chinese or Japanese. “Coverage” is the share of those records carrying a surname on the list for the subgroup, and “Precision” the share of all people carrying a surname on the list, of any race, who belong to the subgroup, both computed from the enumerated race. The estimated coverage is obtained from the county-level list hit rates and the county Asian share alone, by the method of Section 3.

and the Japanese list covers 0.26 of Japanese records with a precision of 0.79; the precision of the Chinese list is low because most people carrying a surname on it are the white Americans described above. As in Appendix A6, we cannot estimate Π without ground-truth labels here, because the estimator of Section 4.3 draws on the subgroup geographic prior. However, the subgroup recovery method of Section 3 identifies the coverage of each list from the county-level list hit rates and the coarse geography alone. The estimated coverage is within 0.03 of its ground truth for both lists, and the recovered county subgroup shares correlate with the truth at 0.97.

Results. We compare the empirical performance of ℓ BISG, which recovers the subgroup geography from the coarse geography and the lists, against oracle ℓ BISG, which uses the true subgroup geography, and against geography alone. As shown in Figure 5, ℓ BISG (red solid line) nearly matches oracle ℓ BISG (blue dashed line): average precision is 0.97 against 0.97 for Chinese records and 0.95 against 0.95 for Japanese records, whereas geography alone yields 0.87 and 0.73.

Figure 5 shows that the predicted subgroup probabilities are calibrated across most of the range under both priors. The expected calibration error is 0.06 under the oracle prior and 0.07 under the recovered prior, and is largest in the lowest bin: among the records assigned a probability near zero of being Chinese, 11% are Chinese, because their surnames appear on neither list and the two subgroups are similar in size.

We also evaluate the validity of downstream analysis based on the probabilities obtained using ℓ BISG. We estimate the share of each subgroup in the top quartile of the 1900 socioeconomic index within each county, among the 138,720 records in the labor force with a recorded score. We fit a single jointly estimated BIRDiE model over the two subgroups with complete pooling (McCartan et al., 2025), and score the counties whose expected number of subgroup members under the recovered prior is at least ten. We compare ℓ BISG against oracle ℓ BISG and geography alone. Figure 6 compares the estimated top-quartile share of each subgroup at the county level against the truth. The bias of ℓ BISG is close to that of oracle ℓ BISG: -0.1 against -0.1 ppts for Chinese records and -1.6 against -1.2 for Japanese, whereas geography alone yields -0.8 and 1.7 . The county RMSE of ℓ BISG is 1.3 against 1.2 ppts for Chinese records and 8.9 against 7.3 for Japanese, against 3.4 and 12.9 for geography alone.

Overall, we find that ℓ BISG, which recovers the subgroup geography from the coarse geography and the lists, matches oracle ℓ BISG in accuracy, calibration, and downstream

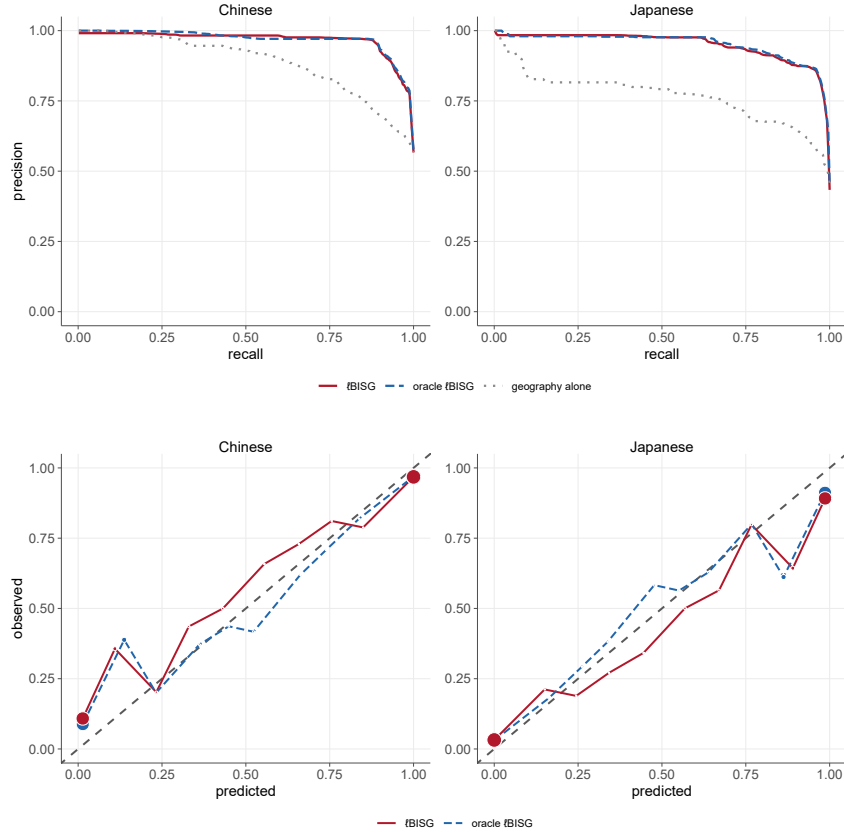


Figure 5: Precision and recall (top) and calibration (bottom) for ℓ BISG, which recovers the subgroup geography from the coarse geography and the lists by the method of Section 3 (red, solid), oracle ℓ BISG, which uses the true subgroup geography (blue, dashed), and, in the top row, geography alone, the county Asian share (grey, dotted), by subgroup. In the bottom row, the dashed diagonal marks perfect agreement between estimated probabilities and observed proportions, and the area of each circle is proportional to the number of records in the bin.

analysis validity. The results depend on the lists being exclusive to their target groups. This validation demonstrates the application of ℓ BISG to subgroups for which neither a surname table nor a geographic prior exists.

5.3 Predicting religious sects in Lebanon using voter files

Finally, we turn to Lebanon and apply ℓ BISG outside the United States to predict individuals' religious sect memberships rather than their racial and ethnic identities.

Setup. We utilize full-count voter files, containing 3,676,322 registrants, which was published by the Lebanese Ministry of Interior and Municipalities through the Directorate General of Civil Status on 1 February 2018. We drop the 14,388 voters whose sectarian memberships are missing. Each file contains the full name, location of residence, and self-reported religious sect of each registrant (Dasanaïke, 2026). We consider the seven largest

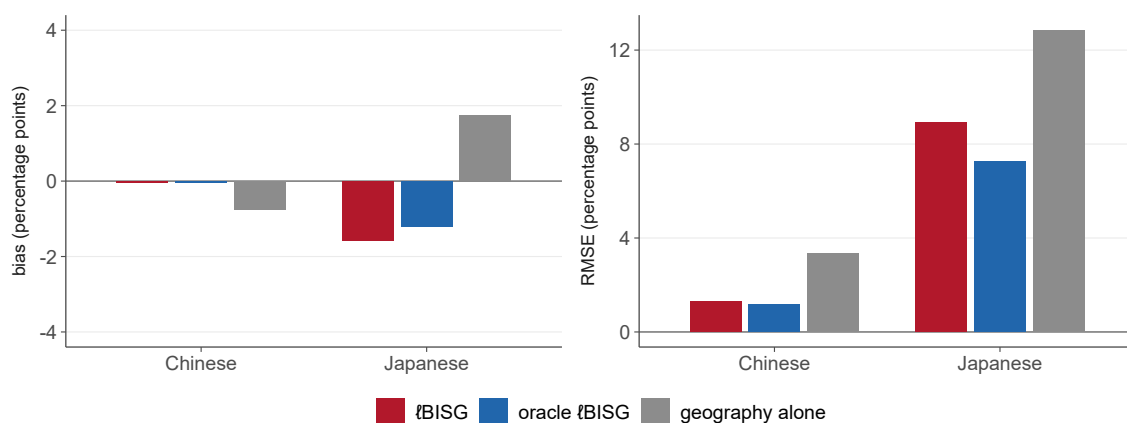


Figure 6: County-level subgroup-specific estimates of the share in the top quartile of the 1900 socioeconomic index. Panels give signed bias (left) and root-mean-square error (right), both in percentage points. Red bars are ℓ BISG, blue bars oracle ℓ BISG, and grey bars geography alone.

sects, together comprising 96.6% of all registrants, and collapse the remaining 125,534 registrants into the residual group. These seven sects are the Shia, at 29.9% of all registrants, Sunni (27.1%), Maronite (19.8%), Roman Orthodox (7.0%), Druze (5.5%), Roman Catholic (4.9%), and Armenian Orthodox (2.4%). No other sect comprises more than 1% of the roll.

Surnames are more informative about religious sects in Lebanon than about racial groups in the United States, because families tend to be confessionally endogamous. However, that informativeness is unevenly distributed. The least distinctive names belong to the smaller Christian sects, whose surnames come from a Levantine Arabic ancestry shared with larger neighbors.

Unfortunately, the last census in Lebanon was conducted in 1932 under French supervision, so the geographic distribution of sect from external sources is not available. Therefore, in this validation exercise, we use the share of each sect among the registrants of the voter’s locality, aggregated from the self-reported sects in the registry itself (there are 1,550 localities), as the geographic prior.

Our downstream analysis uses these locality-level predictions to estimate the age composition of each sect at the district level ($n = 29$). The age composition of Lebanon’s sectarian groups is politically consequential because political representation is explicitly allocated along sectarian lines. Based on the 1932 census, which found Christians to constitute 51% of the population, the 1943 National Pact allocated parliamentary seats to Christians and Muslims in a 55–45 ratio. Following fifteen years of sectarian civil war, the 1989 Taif Agreement revised this allocation to an equal 50–50 split. By that time, however, unofficial estimates placed the Muslim share of the population at roughly 70% (Faour, 2007).

Because younger cohorts are disproportionately Muslim, a proposal to lower the voting age from 21 to 18, endorsed by parliament in 2009, has stalled amid concerns that expanding the electorate to younger voters would place Christian confessions at a substantial numerical disadvantage (Faour, 2007). Although no census has been conducted since 1932, in part to avoid reigniting sectarian conflict over the demographic balance, sectarian affiliation continues to be recorded in the electoral rolls because civil status in Lebanon is administered by confession and parliamentary seats are allocated by sect.

Sect	Registrants	Coverage		Precision	
		Ground truth	Estimate	Ground truth	Estimate
Shia	1,096,570	0.72	0.71	0.48	0.48
Sunni	991,158	0.56	0.58	0.35	0.36
Maronite	724,231	0.75	0.77	0.37	0.38
Roman Orthodox	254,712	0.64	0.68	0.11	0.12
Druze	201,756	0.80	0.83	0.11	0.12
Roman Catholic	180,195	0.61	0.68	0.08	0.08
Armenian Orthodox	87,774	0.24	0.29	0.77	0.93

Table 3: Quality of the seven LLM sect lists on the Lebanese voter registry. “Coverage” is the share of a sect’s registrants carrying a surname on that sect’s list, and “Precision” the share of the registrants carrying a list surname who belong to the sect. Each list holds 1,000 surnames.

Without a published surname table, we construct lists using an approach similar to the one employed in Appendix A6. Specifically, we utilize Gemini 3.0 Flash to propose prompts written in Arabic (see Appendix A7 for details). As done in the previous applications, we make six separate model calls per prompt at temperature 1.0, rank the surnames by the number of calls that returned them, and retain the top 1,000 for each sect.

We embed surnames using the paraphrase-multilingual-mpnet-base-v2 model (Reimers and Gurevych, 2020), which accommodates Arabic script, instead of the E5 encoder used in the previous applications. The list-membership models of Step 2 of Algorithm 1, one per sect list, are fit on a subsample of one million voters drawn at a fixed rate within sect, and we score every registrant. The remaining steps follow Algorithm 1, with $\hat{f}$ estimated as done in the other applications, and $K = 100$ clusters chosen by Algorithm 2 on the full roll. We estimate $\mathbb{P}(B \mid R)$ with two-fold cross-fitting: voters are assigned to two folds at random, and each fold is scored with the estimate obtained from the other.

Quality of name lists. We evaluate the quality of LLM sect lists using an approach similar to the one used in the previous applications. Table 3 reports coverage and precision for each of the seven sects, both as computed from the recorded sects and as estimated from the locality system, using the locality sect shares as the geographic prior and no individual labels (Section 4.3). The estimates align closely to their corresponding ground-truths with the only exception being the estimated precision for Armenian Orthodox, which overestimates precision by 16 ppt.

Across the groups, the Armenian lists yield the lowest coverage but the highest precision, while the Druze and Roman Catholic lists hold surnames that cover most registrants of those sects as well as many non-members. The sect lists also overlap more often than any of the American lists: 57.0% of registrants have a surname that belongs to more than one list. Our overall list quality measure is $\gamma = 0.37$, which is similar to its ground truth 0.33 based on individual sectarian memberships, suggesting that the Lebanese lists are similar in quality to the American lists.

Results. We compare the empirical performance of ℓ BISG based on the LLM lists against geography alone, because no surname frequency table exists for Lebanon from which to

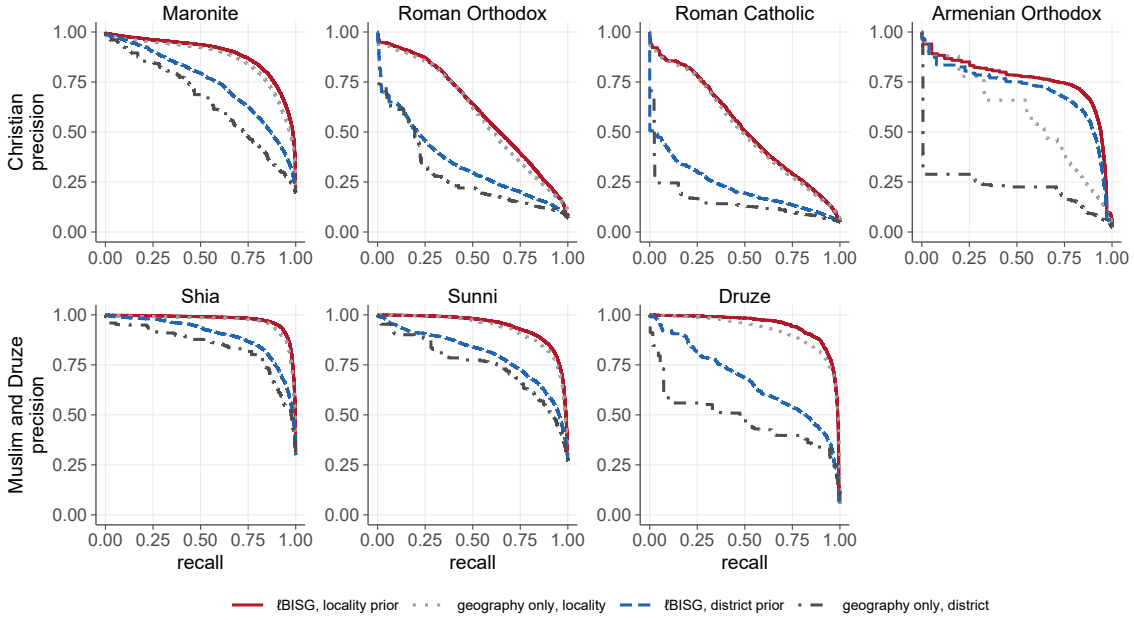


Figure 7: Precision and recall for ℓ BISG and geography alone for the seven principal Lebanese sects, by confession (rows) and sect (columns), under the locality-level prior ($n = 1,550$; ℓ BISG red solid, geography grey dotted) and the district-level prior ($n = 29$; ℓ BISG blue dashed, geography dark grey dot-dashed).

estimate standard BISG. As shown in Figure 7, locality alone (grey dots) already predicts sect well, since most localities are confessionally homogeneous. ℓ BISG (red solid line) improves on locality-alone prediction for every sect, though only marginally for Roman Catholics, and by the most for the Armenian Orthodox, whose average precision rises from 0.48 under geography alone to 0.74; the Armenian Orthodox list is the most precise of the seven (Table 3).

The Muslim sects and the Druze are predicted best, with average precision above 0.94, and the smaller Christian sects worst, at 0.52 for Roman Catholics and 0.61 for Roman Orthodox, because their surnames are shared with the larger Christian sects. A low-precision list still supports precise predictions: the Roman Catholic list has a precision of only 0.08 at a recall equal to its coverage of 0.61 (Table 3), whereas the ℓ BISG posterior reaches a precision of 0.40 at the same recall, since the posterior also draws on geography and on the other six lists.

Among the localities where no single sect comprises more than 90% of the roll (43% of 1,550 localities), however, ℓ BISG exceeds geography alone by between 0.02 (Roman Catholic) and 0.27 (Armenian Orthodox) in average precision, and by 0.05 to 0.09 for the three largest sects. If we use the more heterogeneous district-level ($n = 29$) share as the prior instead of the locality, ℓ BISG (blue dashed line) outperforms district-alone prediction (black dashed line) by between 0.07 and 0.51 in average precision. However, the downstream age estimates in Figure 9 become less accurate, with the RMSE rising for six of the seven sects. This is expected when moving from the sample size of 1,550 localities to 29 districts.

Figure 8 shows that the predicted individual sect probabilities based on ℓ BISG are well

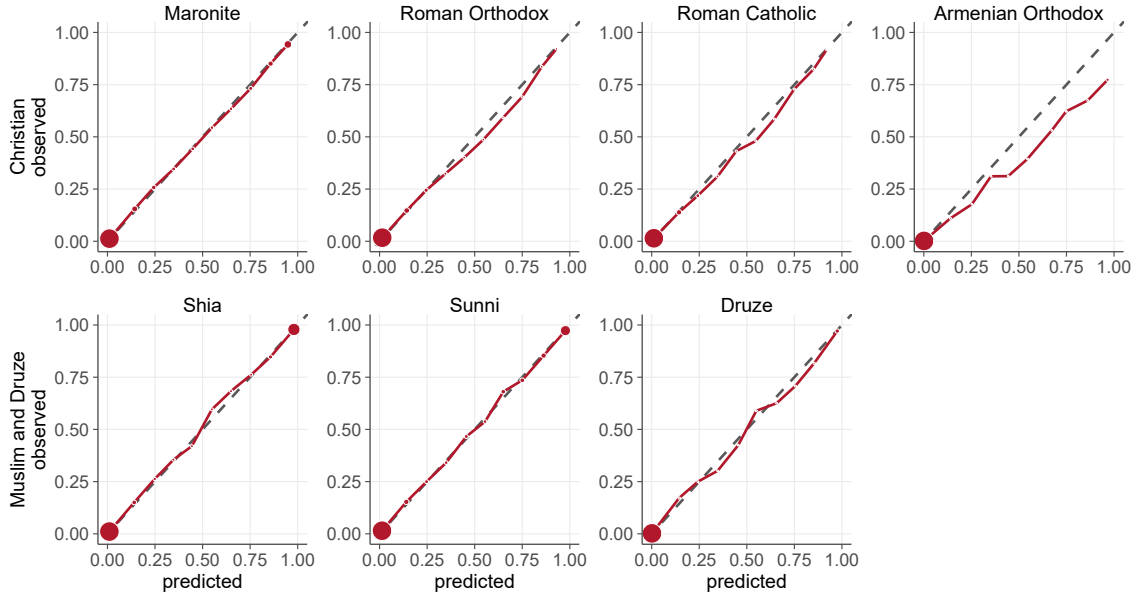


Figure 8: Calibration of estimated individual sect probabilities based on ℓ BISG (LLM lists) for the seven principal Lebanese sects, by confession (rows) and sect (columns). The dashed diagonal marks perfect agreement between estimated probabilities and observed proportions, and the area of each circle is proportional to the number of registrants in the bin.

calibrated for the Maronites, the Muslim sects, and the Druze, whose reliability curves follow the diagonal. The three smaller Christian sects are over-predicted at high probabilities. Among registrants assigned a probability near 0.97 of being Armenian Orthodox, 0.77 are; most of the remainder are Armenian Catholics and Armenian Protestants in the residual class, since the Armenian Orthodox list identifies Armenian surnames rather than the denomination. The expected calibration error is nevertheless low for every sect, from 0.002 for the Druze to 0.006 for the Armenian Orthodox, because most registrants receive a probability near zero for any given sect.

We also evaluate the validity of the downstream analysis based on the estimated individual sect probabilities using ℓ BISG. We measure the share of young registrants in each sect (born in 1985 or later); 90.5% of registrants have a recorded birth year, and we score the sect-by-district pairs with at least 300 registrants. This value varies across the sects, ranging from 10.0% among Armenian Orthodox to 26.8% among Shia. We estimate this share for each sect within districts using the saturated specification of BIRDIE. We evaluate at the district level rather than at the locality level of the prior because most localities are confessionally homogeneous. As in Appendix A6, we compare against geography alone.

Figure 9 shows that across the 157 sect-by-district pairs, ℓ BISG reduces the overall RMSE of the geographic baseline by roughly half, from 13.0 to 6.9 ppts. The absolute bias is smaller for five of the seven sects (top panel), with Sunni essentially unchanged (1.8 to 2.1 ppts) and Roman Catholic worsening slightly from 4.7 to 5.1 ppts. The per-sect RMSE is lower for every sect (bottom panel).

The size of the reduction reflects two separate components: how well geography alone

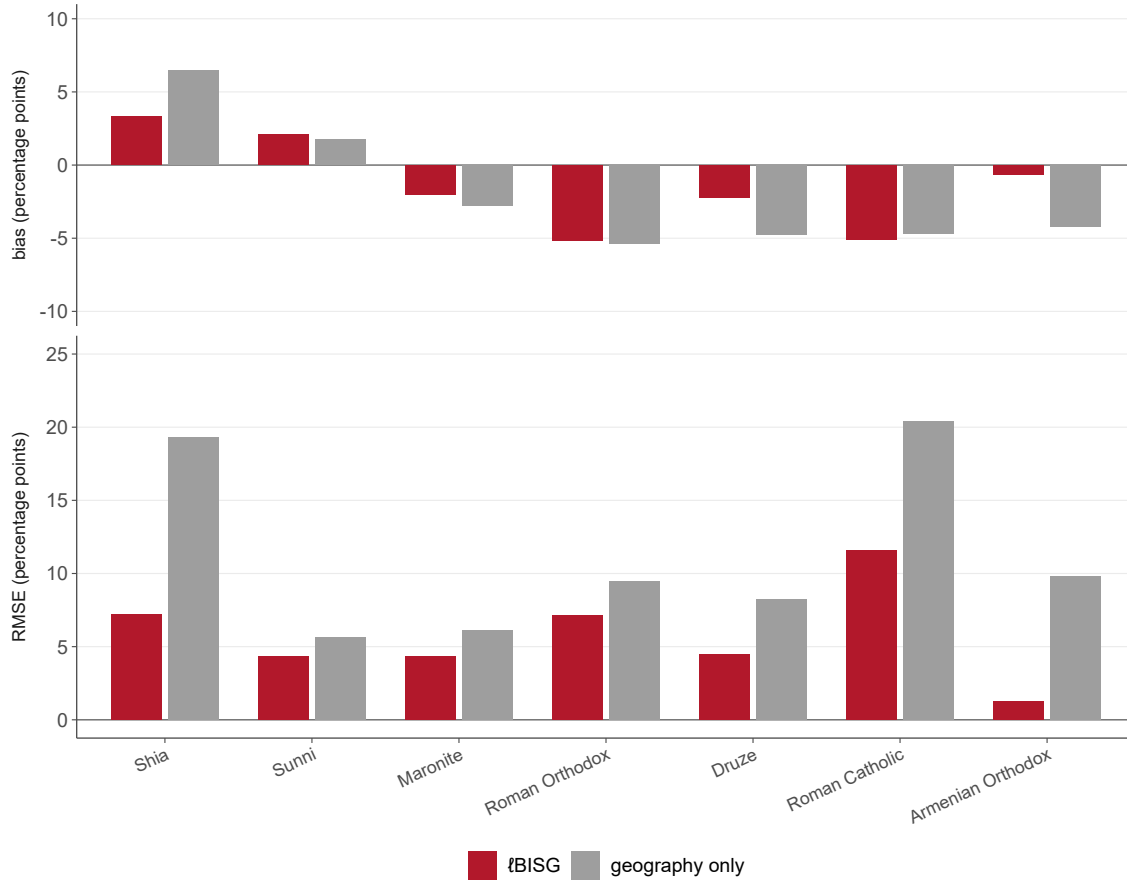


Figure 9: Share of each sect born in 1985 or later, estimated by district. Rows give signed bias and root-mean-square error against the enumerated sect, both in ppts; red bars are ℓ BISG and grey bars geography alone. Sects are ordered by their share of the roll.

recovers the sect due to residential segregation, and how well surnames identify group members. The Roman Catholics, whose members are older and dispersed as a minority across districts, have a baseline RMSE of 20.4 ppts, of which ℓ BISG removes 43%. The Armenian Orthodox, whose list is the most precise (Table 3), have the proportionally largest reduction, from 9.8 to 1.3 ppts. The reduction is smallest for the Sunni, whose age composition is close to that of the districts they live in, so geography alone already produces small errors.

Overall, we find that ℓ BISG predicts sectarian memberships with average precision between 0.52 and 0.98, improves on geography alone for every sect, and roughly halves the error of the geographic baseline on the district-level estimand. This validation demonstrates the potential application of ℓ BISG outside of the United States, to non-English names, and to a non-traditional group membership for which surnames are informative.

6 Concluding Remarks

Bayesian Improved Surname Geocoding (BISG) and its variants have become the standard method for predicting individual race in the US. However, the requirement of race-name frequency tables has severely limited the applicability of BISG beyond this specific context. In particular, these methods were not applicable beyond the coarse racial categories defined by the Census Bureau and outside the US where such information is rarely available.

In this paper, building on our earlier work of embedding-powered BISG (eBISG; [Dasanaike and Imai, 2026](#)), we overcome this difficulty by developing list-powered BISG (ℓ BISG), which only requires group specific name lists, which can be easily obtained from an LLM. Importantly, ℓ BISG does not impose any additional assumptions beyond what is required for the standard BISG method. Under an additional assumption of list distinctiveness, we have also shown how to extend ℓ BISG to the settings, in which the information about subgroup geographic prevalence is not available. Our empirical validations show that the performance of ℓ BISG is comparable to the standard BISG even though the former only requires group specific name lists.

Because one can use a large-language model to generate a name list essentially for any group, ℓ BISG extends probabilistic inference to fine-grained ethnic groups, to settings outside the United States, and to subgroups within a coarse racial group. We find that the ℓ BISG estimates are as accurate as the standard BISG estimates so long as the lists are distinctive. We expect ℓ BISG to considerably widen the applicability of probabilistic prediction methods that estimate group disparities when their membership is not observed.

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Supplementary Appendix

A1 Using the list score directly

Using the Florida data, we show that the list score $f_{r,e} = \mathbb{P}(L_r=1 \mid E = e)$ cannot serve as the race probability on its own. This is because the probability that a name is on the list does not equal the probability that a person with that name belongs to the group.

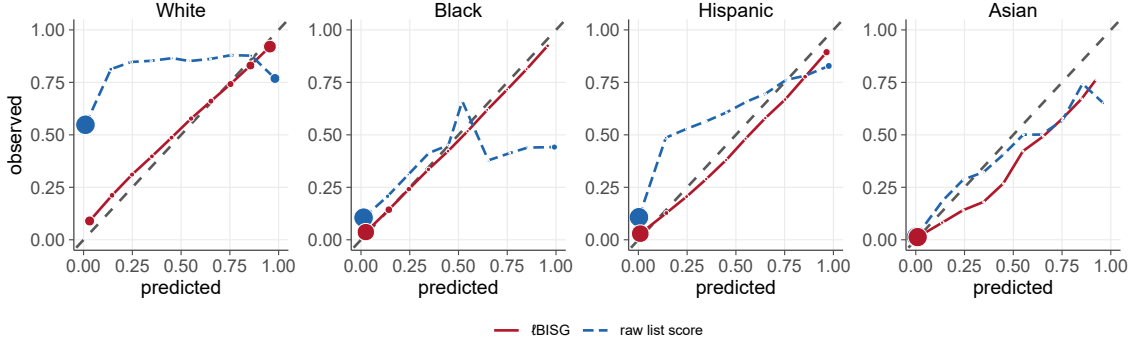


Figure A1: Calibration of the raw list score $f_r = P(L_r=1 \mid E)$ used directly as a race probability (blue, dashed) against ℓ BISG (red, solid), by race, on the Florida registrants. The dashed diagonal marks perfect agreement between estimated probabilities and observed proportions, and the area of each circle is proportional to the number of registrants in the bin.

Figure A1 demonstrates the fact that since lists have imperfect precision and ignore base rates, f (blue dashed line) is not well calibrated. The failure is most dramatic for White, the majority group, whose list is the least distinctive of the four and whose score barely exceeds the base rate. For comparison, Figure A1 also includes the ℓ BISG probabilities, which combine the same score with the geographic prior.

A2 Surname embeddings and the list score

The list score $f_{r,e} = \mathbb{P}(L_r=1 \mid E = e)$ is a smooth function of the name embedding. In Figure A2, we project the Florida surname embeddings to two dimensions. In this plot, groups of similar names form coherent regions, and coloring each embedding point e by the dominant group (i.e., $\arg \max_r f_r$) partitions the space into interpretable regions anchored by familiar names (Figure A2, top). The list scores vary smoothly across that space (Figure A2, bottom). A name that belongs to no list inherits the score of its embedding neighborhood, so it still receives a plausible probability.

Figure A3 highlights that within the Asian group, South Asian surnames stand apart from East and Southeast Asian ones in the embedding, while Chinese, Vietnamese, and Korean surnames overlap. This illustrates the value of setting the number of clusters in the K -means clustering algorithm to a value larger than $d_{\mathcal{R}}$ – we may need to pick up multiple Asian clusters.

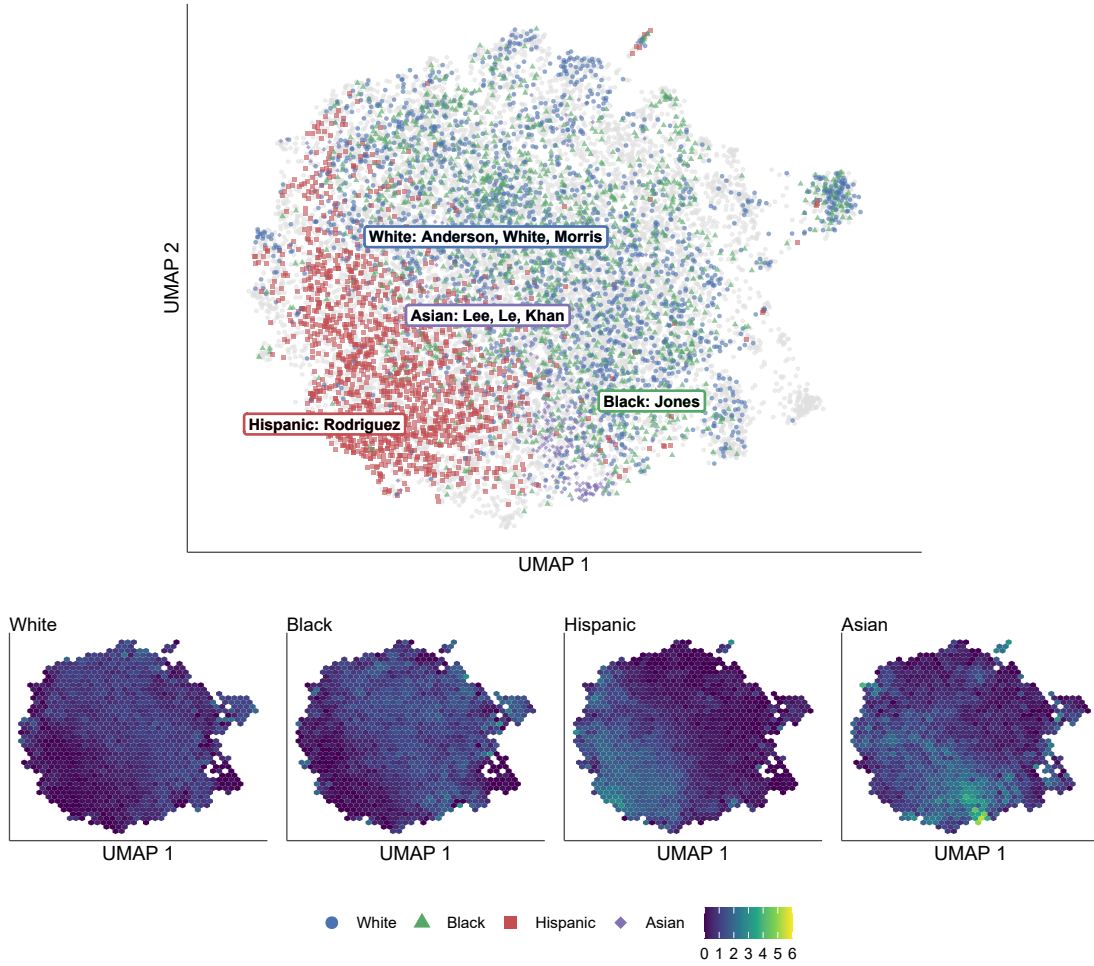


Figure A2: Surname embeddings (E5), Florida. Top: a two-dimensional UMAP projection colored by dominant group (i.e., $\arg \max_r f_r$), with one exemplar label per group. Bottom: the same space, one panel per group, each hex colored by the mean quantile bin of that group's list score $f_r = P(L_r=1 \mid E)$ among the surnames it contains, at different quantiles.

A3 Empirical results for the choice of the number of clusters

Figure A4 compares $Q(K)$ of equation (5) (right panel) with two label-free alternatives, the in-sample entropy of the posterior (left panel), and the entropy of a single held-out fit (middle panel), on the Florida voter file at three sample sizes. The alternatives favor more clusters than $Q(K)$ at every sample size, leading to overfitting. The single-fit held-out entropy selects the largest candidate when the sample is small, because entropy rewards confident predictions regardless of their correctness. The cross-fold agreement in $Q(K)$ penalizes confident but inconsistent predictions produced by overfitting, and $Q(K)$ selects an interior K at every sample size, as does the K that maximizes the ground-truth average precision.

The selection is not sensitive to the partition of geographic units into folds. Across

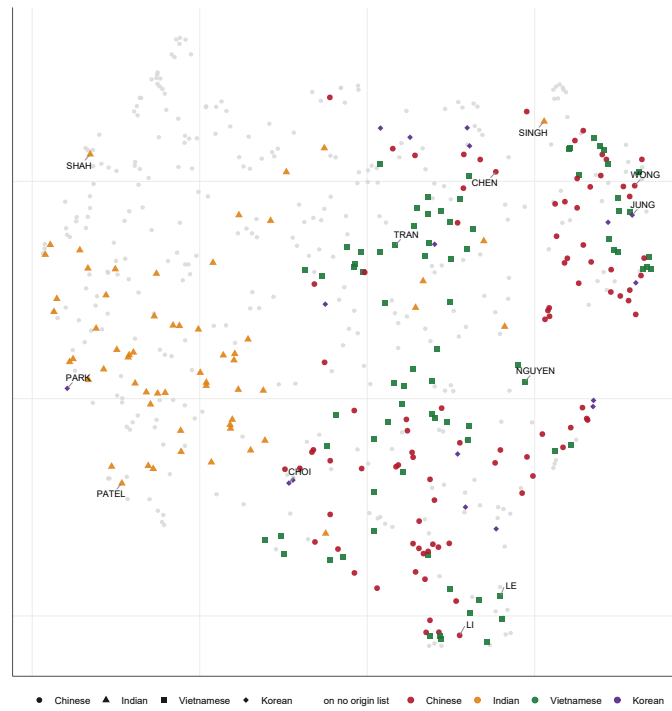


Figure A3: The 587 Asian-dominant Florida surnames, projected on their own and colored by national-origin list. Grey points are on no origin list.

five independent random partitions of the Florida units, the selected K is identical for White, Hispanic, and Asian, returning 75, 150, and 20, respectively, and agrees with the population-balanced partition in each case. For Black, the race where the criterion is flattest, the selection ranges from 30 to 300.

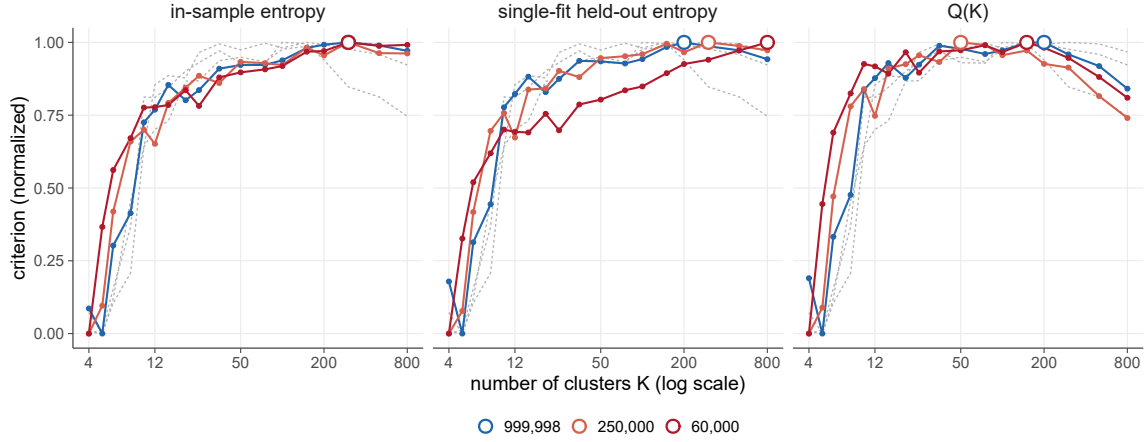


Figure A4: Label-free criteria for choosing the number of clusters K on the Florida voter file, at three sample sizes (colors). Each curve is normalized within its sample size and oriented so that a higher value means the criterion prefers that K ; the open circle marks the K each criterion selects. Left and center: in-sample entropy and single-fit held-out entropy. Right: $Q(K)$ of equation (5). The dashed grey line is the ground-truth average precision for Asian registrants.

A4 Comparison with K-means on embeddings

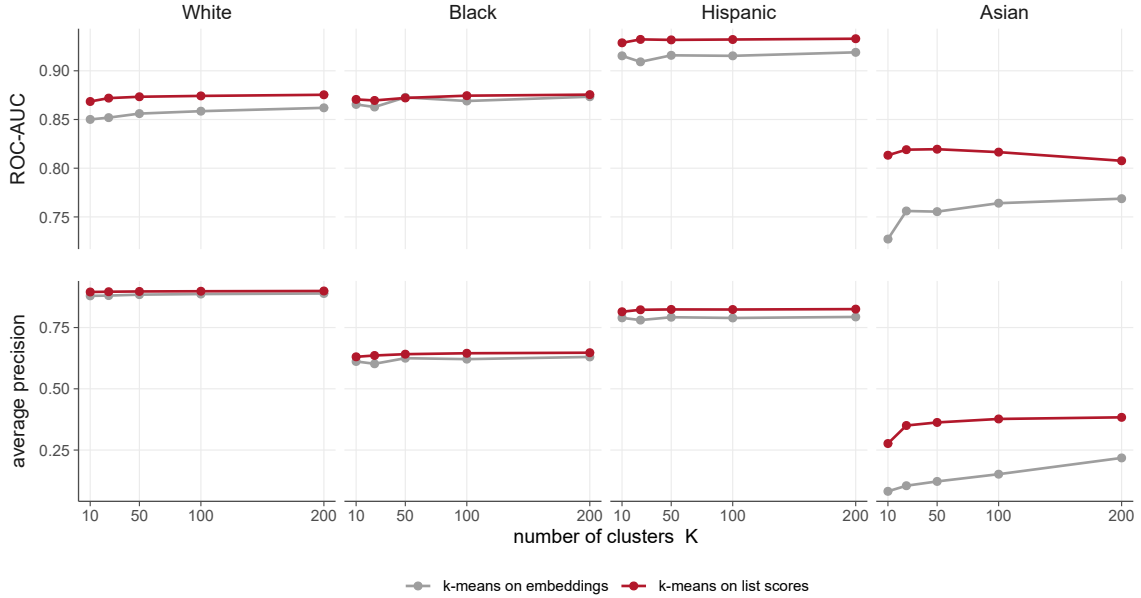


Figure A5: Discrimination when the composite bin is formed by K-means on the raw E5 surname embeddings (no lists) against K-means on the list-score vector f_r , with the same geographic recovery and normalization; Florida, tract-level leave-one-out prior, by cluster count M . Top row: ROC-AUC. Bottom row: average precision.

Figure A5 compares the ROC-AUC and average precision of ℓ BISG under two approaches to the dimension reduction described in Section 2.4. Our proposed approach is K-means clustering on the vectors of log-odds of list scores $f_e = (f_{1,e}, \dots, f_{d_{\mathcal{R}},e})$. We compare this to K-means clustering on the embeddings directly, which relies more heavily on the embeddings being clustered by group membership even if they come from an embedding model trained for other tasks.

In the Florida example, we find that clustering the list scores yields more informative groups than clustering the raw embeddings directly. Discrimination is lower under the raw embeddings for every race and at every cluster count. The difference is substantial for the rare Asian group, whose average precision is 0.10 at twenty-five clusters against 0.35 under the list scores (Figure A5). The original embeddings are therefore not strongly clustered on the Asian group, but the list scores are.

A5 Details for recovery of sub-group geographic probabilities

In this appendix, we provide details of recovering $\mathbb{P}(A = a \mid G = g)$ as discussed in Section 3. We assume assumption 3.1 throughout. Assumption 3.2 is represented by the $\lambda = 0$ case of Section A5.3.

A5.1 Temp brainstorm

What if we are solving not the sub-group problem but the problem that $\mathbb{P}(R = r \mid G)$ are not available? Then, under the perfect separation assumption ($\mathbb{P}(L_{r'} = 1 \mid R = r) = 0$ if $r \neq r'$) and list quality assumption ($L_{r'} \perp\!\!\!\perp G \mid R = r$), we still have a system of equations

$$\begin{aligned}\mathbb{P}(L_{r'} = 1 \mid G) &= \mathbb{P}(L_{r'} = 1 \mid R = r) \mathbb{P}(R = r \mid G) \\ \sum_r \mathbb{P}(R = r \mid G) &= 1\end{aligned}$$

We can still formulate this as a system $\beta = M\Pi$ and $M\mathbf{1}_{d_{\mathcal{R}}} = \mathbf{1}_{d_{\mathcal{G}}}$ where β is the $d_{\mathcal{G}} \times d_{\mathcal{R}}$ matrix of $\mathbb{P}(L_r = 1 \mid G)$, Π is the $d_{\mathcal{R}} \times d_{\mathcal{R}}$ matrix of within group list rates $\mathbb{P}(L_{r'} = 1 \mid R = r)$, and M is again the $d_{\mathcal{G}} \times d_{\mathcal{R}}$ matrix of $\mathbb{P}(R = r \mid G)$. We now want to solve for M . Assuming β is full rank and Π is diagonal (as below), we then can follow the same kinds of algebraic steps starting with

$$\Pi^{-1}\mathbf{1}_{d_{\mathcal{R}}} = (\beta^{\top}\beta)^{-1}\beta^{\top}\mathbf{1}_{d_{\mathcal{G}}}$$

which implies

$$\pi_r = \lambda + \frac{1}{((\beta^{\top}\beta)^{-1}\beta^{\top}\mathbf{1}_{d_{\mathcal{G}}})_r}$$

By setting $\lambda = 0$ and calculating $\mathbb{P}(L_r = 1 \mid G)/\pi_r$, we obtain our estimate of $\mathbb{P}(R = 1 \mid G)$.

Overall: I think we should rework section 3 to be about (1) what if do not have $\mathbb{P}(R = r \mid G)$? The above solution already captures the key ideas involved without bringing in the complication of sub-groups. (2) what if we are interested in sub-groups and only have the coarser groups? Give the slightly more complicated solution while showing how this is just a generalization of problem (1).

A5.2 Formulating the system of equations

Stacking main text equation (7) over geographies and groups gives the following matrix form,

$$\beta = \Theta \Pi, \quad (\text{A1})$$

where β is the $d_G \times d_A$ matrix of list rates $\mathbb{P}(L_a = 1 \mid G)$, Θ is the $d_G \times d_A$ matrix of subgroup geography proportions $\mathbb{P}(A = a \mid G)$, and Π is the $d_A \times d_A$ matrix of within-subgroup list rates $\mathbb{P}(L_{a'} = 1 \mid A = a)$. We also index the A groups and order each of these matrices so that $A = a$ levels belonging to the same $R = r$ appear in blocks.

Next, recall that M is the $d_G \times d_R$ matrix of $\mathbb{P}(R = r \mid G = g)$ and let U be a $d_A \times d_R$ block-diagonal indicator matrix with, in each column r , a vector $\mathbf{1}_{d_A(r)}$ padded by 0's (where $\mathbf{1}_d$ denotes a d -dimensional vector of ones) that encodes the a values such that $r(a) = r$. Then, main text equation (8) can be written as,

$$\Theta U = M. \quad (\text{A2})$$

Under our set-up, both β and M are identifiable. While M is of full rank by Assumption 2.3, we will also assume below that β is of full rank. List prevalence must therefore vary sufficiently by geography. Our goal is to recover Θ , but the key complication is that Π is also unknown.

As discussed in the main text, one way to address the lack of identification is to make assumption 3.2, which imposes that Π is diagonal. In the next section, we generalize this to assuming that Π has a constant off-diagonal value $\lambda \in [0, 1]$ while its diagonal is free to vary. We show that this is sufficient to make Π and therefore Θ uniquely identified. Our main analysis is the special case with $\lambda = 0$, and the result this can be used for sensitivity analysis by setting λ to different small values and seeing how much it changes the final estimates.

A5.3 Sensitivity analysis under uniform cross-group probabilities

Suppose each off-diagonal entry of Π is set to λ , representing the probability that the name of a member of group $A = a$ appears on the list associated with a different group $a' \neq a$. To start, we assume that this probability is constant across all pairs of distinct groups, regardless of whether they are in the same coarse group or not. Thus, λ captures the degree of “error” or lack of separation among the lists: if the lists perfectly separate the groups, then $\lambda = 0$ (assumption 3.2). We leave the diagonal entries of Π unrestricted because they represent the coverage of each list within its corresponding group. Formally, our additional identification assumptions are given below.

Assumption A5.1 (List separation). Let Π be the $d_A \times d_A$ matrix of $\pi_{a',a} = \mathbb{P}(L_{a'} = 1 \mid A = a)$ with diagonal $\pi_a = \pi_{a,a}$. Let β be the $d_G \times d_A$ matrix of list geographic rates $\mathbb{P}(L_a = 1 \mid G)$. Assume:

1. β is full column rank
2. Π is invertible
3. $\Pi = D + \lambda \mathbf{1}_{d_A} \mathbf{1}_{d_A}^\top$ where $D = \text{diag}(d_1, \dots, d_{d_A})$ with $d_a = \pi_a - \lambda$ and $\lambda \in [0, 1]$.

Under these assumptions, equations (A1) and (A2) imply that by regressing each column of M on β , we can recover $\Pi^{-1}U$,

$$\Theta U = \beta \Pi^{-1} U = M$$

$$\iff (\beta^\top \beta)^{-1} \beta^\top M = (D + \lambda \mathbf{1}_{d_A} \mathbf{1}_{d_A}^\top)^{-1} U = \left(1 + \lambda \sum_a \frac{1}{d_a}\right)^{-1} D^{-1} U$$

where the last equality follows from the Sherman-Morrison formula. Together with the fact that $\pi_a = d_a + \lambda$, we obtain,

$$\pi_a = \lambda + \left(1 + \lambda \frac{\mathbf{1}_{d_A}^\top (\beta^\top \beta)^{-1} \beta^\top M \mathbf{1}_{d_A}}{1 - \lambda \mathbf{1}_{d_A}^\top (\beta^\top \beta)^{-1} \beta^\top M \mathbf{1}_{d_A}}\right)^{-1} \left\{((\beta^\top \beta)^{-1} \beta^\top M)_{a, r(a)}\right\}^{-1}. \quad (\text{A3})$$

Full proof of this equation is given in Section A5.5 below.

Alternatively, we can relax assumption A5.1 to allow Π to have different cross-group-rates in different coarse groups r . Specifically, suppose we relax part 3 of this assumption to

$$\mathbb{P}(L_a = 1 \mid A = a) = \begin{cases} \lambda & \text{if } a \neq a' \text{ and } r(a) = r(a') \\ \gamma & \text{if } a \neq a' \text{ and } r(a) \neq r(a') \end{cases}$$

This says that the chance of a member of group a having their name on the list of another sub-group within the same coarser group is λ while the chance of their name being on a list for a sub-group of a different coarser group is γ . Usually, we would expect $\gamma < \lambda$. We might even be willing to assume $\gamma = 0$ so that the coarse groups are separated even if the sub-groups are not.

In matrix form, this new Π can be written

$$\Pi = D + UC U^\top$$

where

$$C = (\lambda - \gamma) I_{d_R} + \gamma \mathbf{1}_{d_R} \mathbf{1}_{d_R}^\top \quad (\text{A4})$$

Under this specification of Π , we prove two results (see Section A5.5). First, if $\gamma = 0$, then Π has a block diagonal form and all relevant equations are completely separable by r . We then simply have equation (A3) within each r group. That is, for sub-group a from course group r ,

$$\pi_a = \lambda + \left(1 + \lambda \frac{\mathbf{1}_{d_A(r)}^\top (\beta_r^\top \beta_r)^{-1} \beta_r^\top M_r}{1 - \lambda \mathbf{1}_{d_A(r)}^\top (\beta_r^\top \beta_r)^{-1} \beta_r^\top M_r}\right)^{-1} \left\{((\beta_r^\top \beta_r)^{-1} \beta_r^\top M_r)_a\right\}^{-1}. \quad (\text{A5})$$

where β_r is the $d_G \times d_A(r)$ matrix of $\mathbb{P}(L_a = 1 \mid G)$ for $a : r(a) = r$; where M_r is the vector of $\mathbb{P}(R = r \mid G = g)$ over g ; and where we are solving for elements of Π_r , the $d_A(r) \times d_A(r)$ matrix of $\mathbb{P}(L_{a'} = 1 \mid A = a)$ for a, a' in the same coarse group.

More generally, for $\gamma > 0$ and $\gamma \neq \lambda$, we get

$$\pi_a = \lambda + \frac{(I_{d_R} - C U^\top (\beta^\top \beta)^{-1} \beta^\top M)_{r,j}}{((\beta^\top \beta)^{-1} \beta^\top M)_{a,j}} \quad (\text{A6})$$

for any j such that $((\beta^\top \beta)^{-1} \beta^\top M)_{a,j} \neq 0$. Note that the $\gamma = \lambda > 0$ case is given in equation (A3).

A5.4 Implementation details under $\lambda = 0$

To estimate $\mathbb{P}(A = a \mid G = g)$ in practice, we use non-negative least squares fit separately for each coarse group (as discussed above, the $\lambda = 0$ case above allows us to partition Π into a block for each R). Let $\rho_a(g) = \mathbb{P}(L_a = 1 \mid G = g)$ and recall that $\pi_a = \mathbb{P}(L_a = 1 \mid A = a)$. From above, we know that

$$\sum_{a:r(a)=r} \rho_a(g) = \mathbb{P}(R = r \mid G = g) \quad (\text{A7})$$

For each r , we solve

$$\hat{\pi} = \arg \min_{\pi \geq 0} \sum_g n_g \left(\sum_{a:r(a)=r} \pi_a \rho_a(g) - \mathbb{P}(R = r \mid G = g) \right)^2,$$

where $\pi \geq 0$ is a shorthand for all its elements being non-negative. Here, we weight each geography by the number of units in that geography. Hence this optimization problem is equivalent to, for each unit i in the dataset, constructing a pair

$$\rho_i := (\rho_1(G_i), \dots, \rho_{d_A(r)}(G_i))^\top, \quad \mathbb{P}(R_i = r \mid G_i) \quad (\text{A8})$$

and running the regression of $\mathbb{P}(R_i = r \mid G_i)$ on ρ_i subject to the coefficients of ρ_i being non-negative. We then set

$$\hat{\theta}_a(g) = \frac{\rho_a(g)}{\hat{\pi}_a} \quad (\text{A9})$$

Thought: should we mention the n_g weighting in the main text? Otherwise this is slightly different from what we imply there?

Kyla comment to Noah: I don't understand the below and it does not seem in line with the theory. Is it perhaps out-dated? Above, I've written my understanding of what this should look like. Could you explain any differences between what you do and what I have above? Regarding the below, specifically: (1) $\sum_a \rho_a(g) / d_a = 1$ does not make sense to me. This is equivalent to saying $\sum_a \rho_a(g) = \pi_a$ since $d_a = \pi_a - \lambda$ and $\lambda = 0$ here. (2) you say n_g is number of coarse group members but we are not assuming we know who the coarse group members are – i.e. R is still unknown. Hence I am assuming you mean just the number of units in geography g ?

To estimate $\mathbb{P}(A = a \mid G = g)$ in practice, we use non-negative least squares. Let $\rho_a(g) = \mathbb{P}(L_a = 1 \mid G = g)$ be the share of the population in geography g that carries a name on list a . In addition, Since the subgroups partition the coarse group, we have $\sum_a \rho_a(g) / d_a = 1$ for every g . We solve

$$\hat{c} = \arg \min_{c \geq 0} \sum_g n_g \left(\sum_a c_a \rho_a(g) - 1 \right)^2,$$

and set $\hat{\theta}_a(g) = \hat{c}_a \rho_a(g)$, rescaled to sum to one within each geography.

We weight each geography by the number of coarse group members that it contains n_g . The constraint $c \geq 0$ prevents negative shares, and the rescaling makes the shares within each geography sum to one. The recovered coverage of list a is given by $1/\hat{c}_a$.

A5.5 Proofs

Proof of equation (A3). Start with the system of equations $\beta = \Theta \Pi$ and $\Theta U = M$. Under the invertibility of Π and β from assumption A5.1, these imply

$$\beta \Pi^{-1} U = M \iff \Pi^{-1} U = (\beta^\top \beta)^{-1} \beta^\top M$$

Substituting in the simplified form of Π under assumption A5.1, this becomes:

$$(D + \lambda \mathbf{1}_{d_A} \mathbf{1}_{d_A}^\top)^{-1} U = (\beta^\top \beta)^{-1} \beta^\top M$$

Next, by the Sherman-Morrison formula,

$$\begin{aligned} \Pi^{-1} &= (D + \lambda \mathbf{1}_{d_A} \mathbf{1}_{d_A}^\top)^{-1} \\ &= D^{-1} - \frac{\lambda (\mathbf{1}_{d_A}^\top D^{-1} \mathbf{1}_{d_A}) D^{-1}}{1 + \lambda (\mathbf{1}_{d_A}^\top D^{-1} \mathbf{1}_{d_A})} \\ &= \left(1 - \frac{(\sum_{a=1}^{d_A} \frac{\lambda}{d_a})}{1 + (\sum_{a=1}^{d_A} \frac{\lambda}{d_a})} \right) D^{-1} \\ &= \left(1 + \lambda \sum_{a=1}^{d_A} \frac{1}{d_a} \right)^{-1} D^{-1} \\ \Pi^{-1} U &= \left(1 + \lambda \sum_{a=1}^{d_A} \frac{1}{d_a} \right)^{-1} D^{-1} U \end{aligned}$$

where $D^{-1} U$ has the same block-diagonal form as U but with, in each column r , the $\mathbf{1}_{d_A(r)}$ vector replaced by the vector of values $\frac{1}{d_a}$ for the a such that $r(a) = r$. Putting everything together, this says that for each a from group $r = r(a)$,

$$\frac{1}{d_a} \left(1 + \lambda \sum_{a=1}^{d_A} \frac{1}{d_a} \right)^{-1} = ((\beta^\top \beta)^{-1} \beta^\top M)_{a,r}$$

Finally, we can solve for $\sum_{a=1}^{d_A} \frac{1}{d_a}$ using the fact that $\mathbf{1}_{d_A}^\top \Pi^{-1} U \mathbf{1}_{d_A}$ is calculable. Specifically, the above equalities imply

$$\mathbf{1}_{d_A}^\top (\beta^\top \beta)^{-1} \beta^\top M \mathbf{1}_{d_A} = \mathbf{1}_{d_A}^\top \Pi^{-1} U \mathbf{1}_{d_A} = \left(1 + \lambda \sum_{a=1}^{d_A} \frac{1}{d_a} \right)^{-1} \sum_{a=1}^{d_A} \frac{1}{d_a}$$

Which, rearranged, implies

$$\sum_{a=1}^{d_A} \frac{1}{d_a} = \frac{\mathbf{1}_{d_A}^\top (\beta^\top \beta)^{-1} \beta^\top M \mathbf{1}_{d_A}}{1 - \lambda \mathbf{1}_{d_A}^\top (\beta^\top \beta)^{-1} \beta^\top M \mathbf{1}_{d_A}}$$

Finally, putting everything together and using $d_a = \pi_a - \lambda$,

$$\pi_a = \lambda + \left(1 + \lambda \frac{\mathbf{1}_{d_A}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} \mathbf{1}_{d_A}}{1 - \lambda \mathbf{1}_{d_A}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} \mathbf{1}_{d_A}} \right)^{-1} \left\{ ((\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{a,r(a)} \right\}^{-1}.$$

Note that setting $\lambda = 0$ simplifies the above derivation. In that case, $\boldsymbol{\Pi}$ is already the diagonal matrix $\mathbf{D}$ and the expressions for $\frac{1}{d_a}$ can be obtained immediately from $(\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M}$. That is,

$$d_a = \{((\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{a,r(a)}\}^{-1}$$

which is the reciprocal of an ordinary least squares coefficient. $\square$

Proof of equation (A5). Suppose $\gamma = 0$. Then $\boldsymbol{\Pi}$ consists of blocks for each coarse group which each takes exactly the form of the overall $\boldsymbol{\Pi}$ in the previous calculations. Moreover, each $\sum_{a:r(a)=r} \mathbb{P}(A = a \mid G) = \mathbb{P}(R = r \mid G)$ is entirely separate from those for other r' , so the entire system of constraints can be solved only within each coarse group block. Hence the result follows immediately from doing the calculations above within each coarse group block. $\square$

Proof of equation (A6). Suppose $\gamma > 0$ and $\gamma \neq \lambda$. We again start with

$$(\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} = \boldsymbol{\Pi}^{-1} \mathbf{U}$$

Rearranging this equation and substituting in the form of $\boldsymbol{\Pi}$ yields

$$\begin{aligned} \mathbf{U} &= \boldsymbol{\Pi} (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} \\ &= (\mathbf{D} + \mathbf{U} \mathbf{C} \mathbf{U}^\top) (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} \\ &= \mathbf{D} (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} + \mathbf{U} \mathbf{C} \mathbf{U}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} \\ \mathbf{D} (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} &= \mathbf{U} (\mathbf{I}_{d_R} - \mathbf{C} \mathbf{U}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M}) \end{aligned}$$

The right side of this equation consists of entirely known quantities. The left side includes multiplication by a diagonal matrix $\mathbf{D}$. Consider the a^{th} row with $r = r(a)$. Then for any column j of the left matrix, we have

$$\begin{aligned} d_a \left((\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M} \right)_{a,j} &= e_r^\top (\mathbf{I}_{d_R} - \mathbf{C} \mathbf{U}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{\cdot,j} \\ &= (\mathbf{I}_{d_R} - \mathbf{C} \mathbf{U}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{r,j} \end{aligned}$$

As long as the left a, j entry is non-zero, and using $d_a = \pi_a - \lambda$ this can be rearranged to

$$\pi_a = \lambda + \frac{(\mathbf{I}_{d_R} - \mathbf{C} \mathbf{U}^\top (\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{r,j}}{((\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{a,j}} \quad (\text{A10})$$

for any j such that $((\boldsymbol{\beta}^\top \boldsymbol{\beta})^{-1} \boldsymbol{\beta}^\top \mathbf{M})_{a,j} \neq 0$. $\square$

A6 Predicting national origin using US historical census

We apply ℓ BISG to the 1900 census of Section 5.2 a second time, to predict the national origin of the white population. The census records the country birthplaces of both parents, and we validate ℓ BISG estimates against the enumerated country birthplace of each person's father. Unlike the Asian application of Section 5.2, the subgroup surnames in this application are also held by people outside the coarse group, so this application requires an additional step, described below, before we apply the recovery of Section 3.

Setup. We use all 75,788,059 records across 2,815 counties, and use the country of birth reported for the father, from whom the surname is typically inherited, as the quantity of interest.

As the coarse group, we use the white population whose father was born abroad, comprising 22.6% of all records, or 24,387,774 people. The published tables of the 1900 census report the size of this group in every county, together with the size of the native white population of native parentage (63.6%) and of the population of any other race (13.7%), but do not report the population by country of origin. As in Section 5.2, the list hit rates used in the recovery are computed over the entire population of each county. The downstream analysis again estimates each bloc's socioeconomic standing at the county level.

The additional assumption for the current setting, in which the geographic prior is available only for the coarse group, is that a person has a surname on a bloc's list only if their father was born in that bloc (see Assumption A5.1). To make this assumption credible, we define the target subgroups as the following four country blocs of origin based on language-of-origin families; Germanic (Germany, Austria, Switzerland, and the Netherlands), Scandinavian (Sweden, Norway, Denmark, and Finland), West Slavic (Poland and Bohemia), and Italian, together with a residual class comprising every other foreign white birthplace.

This assumption fails when the target subgroups are defined at the country rather than the bloc level, because of shared surnames between countries (Austrians with German surnames, Danes with Swedish ones, and so on). It also fails outside the coarse group: white Americans of native parentage are descended from the same immigrant groups and carry the same surnames, so a surname on the Germanic list is held by many people whose father was born in the United States. Because the list hit rates are computed over the entire population of each county, these people are included in every hit rate, and the recovery of Section 3 treats them as members of the coarse group. Among all people carrying a bloc surname, the share who are white of native parentage is 0.09 for the Italian list, 0.17 for West Slavic, 0.26 for Scandinavian, and 0.43 for Germanic; for the residual list it is 0.74.

We therefore estimate, for each list, its hit rate among the two other coarse groups and drop that part of every hit rate before the recovery. In the 1,039 counties where the white population of foreign parentage is less than 5% of residents, nearly every person carrying a list surname belongs to one of the other two groups, so regressing the hit rate of the list on the share of the native white population of native parentage and the share of the population of any other race in those counties, with no constant term, identifies the two rates. Dropping the predicted part from the hit rate of every county leaves the part that belongs to the coarse group, to which we apply the recovery of Section 3. This step requires only the published

Bloc	Father’s birthplace	Records	Coverage		Precision
			Ground truth	Estimate	Ground truth
Germanic	Germany, Austria, Switzerland, Netherlands	8,668,223	0.09	0.15	0.87
Scandinavian	Sweden, Norway, Denmark, Finland	2,309,225	0.16	0.19	0.75
West Slavic	Poland, Bohemia	1,071,200	0.05	0.08	0.73
Italian	Italy	745,671	0.07	0.05	0.88
Residual	all other countries	11,593,455	0.48	0.57	0.84

Table A1: Quality of the four bloc lists and the residual list on the white population of foreign parentage in the full-count 1900 census. “Records” counts the people whose father was born in the bloc’s countries. “Coverage” is the share of those records carrying a surname on the bloc’s list, and “Precision” the share of people carrying a list surname whose father was born there, both computed from the enumerated birthplaces.

coarse shares and a set of counties where few members of the coarse group live; it uses no individual labels.

As in Section 5.2, oracle ℓ BISG uses the county geography of the four blocs. Under the second and more realistic setting, we treat only the three published coarse shares as known (white of foreign parentage, white of native parentage, and any other race). We recover the bloc shares in each county from the list hit rates using the method described in Section 3 (assuming $\lambda = 0$), after the additional step described next.

We construct a list for each separate country of origin by the procedure of Section 5.2, and define the list for each bloc as the union of its member origins’ lists. For Germany, for example, the model-proposed descriptions are “German immigrant families in the rural US Midwest (ordinary farmers, not famous people)”, “German immigrant families (Prussian, Bavarian, Saxon origins) in 19th-century America”, and “German-American families in the Great Plains and Wisconsin/Minnesota.”

The residual class also requires a list, which we construct from two sources: a list of common surnames of white Americans of native parentage, generated by the same LLM procedure with the meta-prompt phrased for that population (“the white American population of native parentage around 1900”), and the lists of the countries assigned to the residual. Appendix A7 lists the descriptions generated for every group.

We then apply the exclusivity filter across all five bloc lists, dropping every surname that appears on more than one. Before applying the exclusivity filter, 3.9% of the records had a surname on more than one bloc list. After filtering, the bloc lists hold between 859 (Italian) and 3,297 (Scandinavian) surnames, and the residual list 7,526.

The remaining steps follow Section 5.2. Algorithm 2 returns $K = 900$ under the oracle prior and $K = 3,000$ under the recovered prior, with cross-fitting by individual records.

Quality of name lists. Table A1 reports coverage and precision for each bloc list computed from the ground-truth birthplaces. Coverage is low for every bloc, ranging from 0.05 for West Slavic to 0.16 for Scandinavian, in part because the exclusivity filter leaves relatively short lists. Precision, however, is universally high, ranging from 0.73 for West Slavic to 0.88

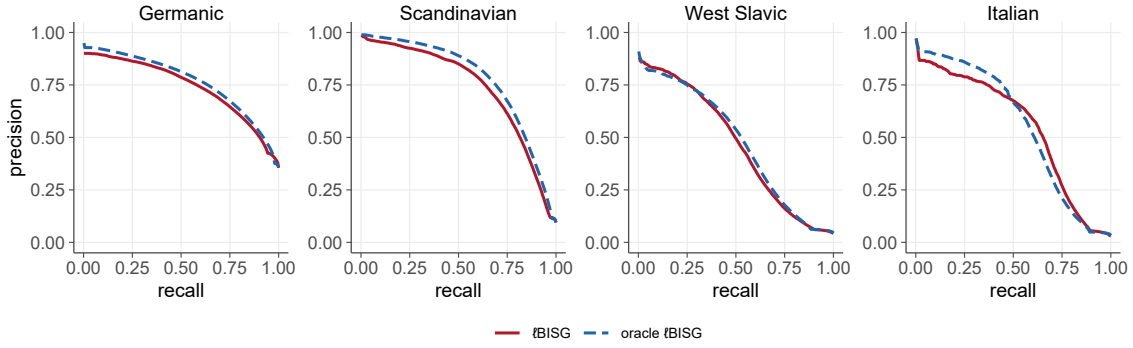


Figure A6: Precision and recall for ℓ BISG, which recovers the bloc geography from the coarse geography and the lists by the method of Section 3 (red, solid), and oracle ℓ BISG, which uses the true bloc geography (blue, dashed), by bloc.

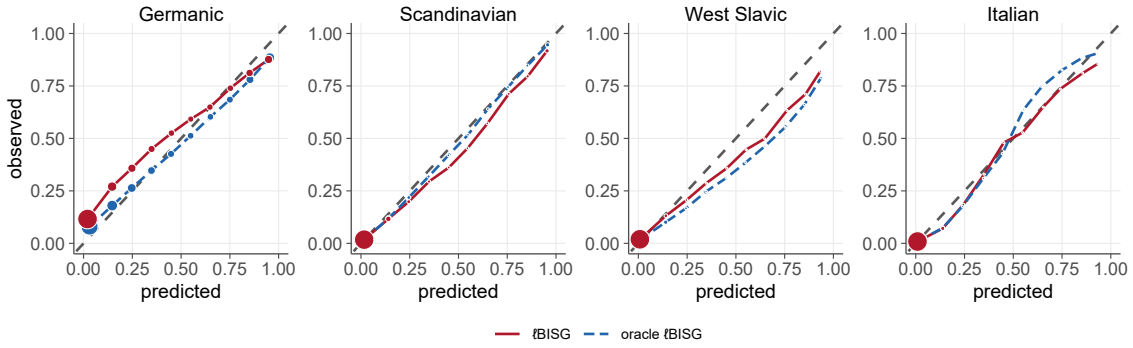


Figure A7: Calibration of estimated individual bloc probabilities based on ℓ BISG (red, solid) and oracle ℓ BISG (blue, dashed), by bloc. The dashed diagonal marks perfect agreement between estimated probabilities and observed proportions, and the area of each circle is proportional to the number of records in the bin.

for Italian. The list quality measure γ is 0.16, which is lower than in Florida (see Section 5.1) because the filtered lists are short, so the diagonal entries of Π are small even though the off-diagonal entries are near zero.

As in Section 5.2, we cannot estimate Π without ground-truth labels here, but the subgroup recovery identifies the coverage of each list. The estimated coverage is within 0.06 of its ground truth for every bloc, and within 0.09 for the residual list.

Results. We compare the empirical performance of ℓ BISG, which recovers the bloc subgroup geography from the coarse geography and the lists, against oracle ℓ BISG, which uses the true bloc geography. As shown in Figure A6, ℓ BISG (red solid line) approaches oracle ℓ BISG (blue dashed line) for every bloc: average precision is 0.42 against 0.56 for Italian records, 0.73 against 0.78 for Scandinavian, 0.69 against 0.77 for Germanic, and 0.44 against 0.48 for West Slavic. Discrimination is highest for Scandinavian and Germanic records and lowest for West Slavic records, whose list has the lowest precision (see Table A1). The

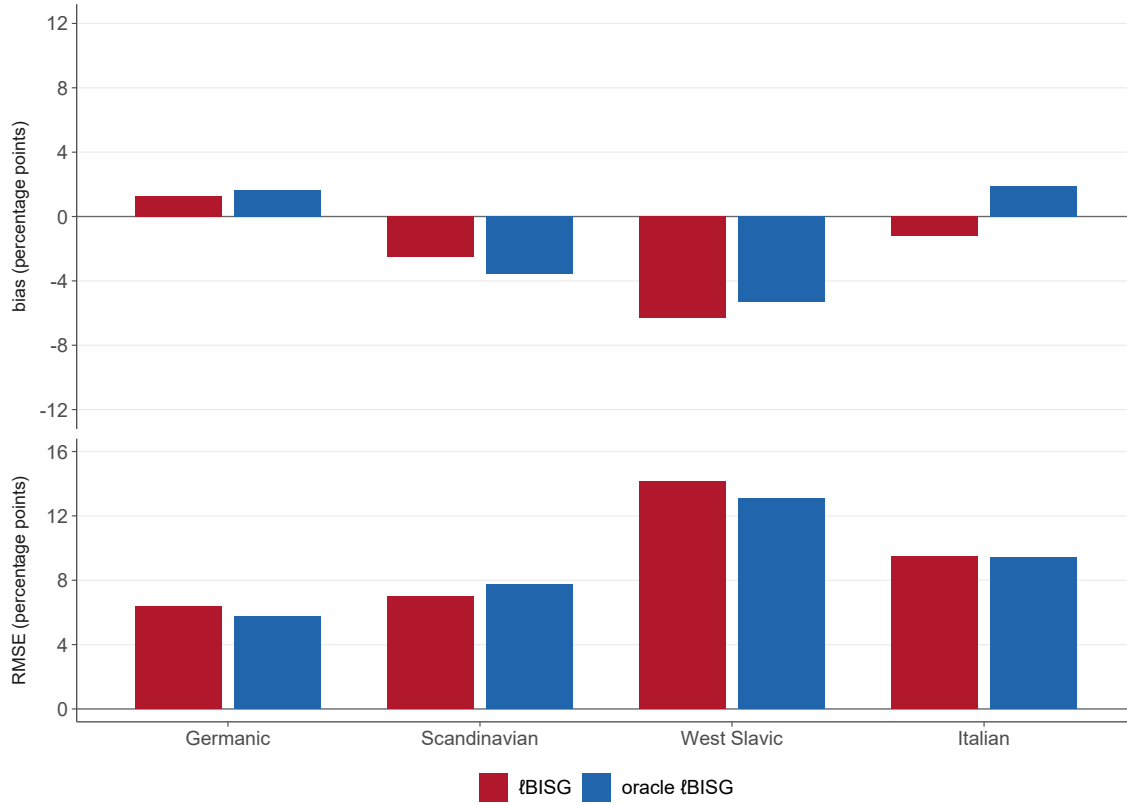


Figure A8: County-level bloc-specific estimates of the share in the top quartile of the 1900 socioeconomic index. Rows give signed bias (top) and root-mean-square error (bottom), both in percentage points. Red bars are ℓ BISG and blue bars oracle ℓ BISG. Blocs are ordered by their share of the coarse group.

recovered county shares correlate with the truth at between 0.82 and 0.95 across the four blocs, and the recovered share of the residual class is 0.44 against a truth of 0.48.

Figure A7 shows that the predicted bloc probabilities are calibrated under oracle ℓ BISG and, less closely, under ℓ BISG. The expected calibration error is below 0.02 for three of the four blocs under oracle ℓ BISG, and 0.044 for Germanic, and between 0.017 (West Slavic) and 0.038 (Germanic) under ℓ BISG. The recovery overstates the Italian share of the coarse group, at 0.06 against a ground truth of 0.03, so ℓ BISG assigns Italian records probabilities above their observed frequency. Nevertheless, these results show that the ℓ BISG method yields reasonably accurate and calibrated individual probabilities of ethnic group memberships.

We estimate the share of each bloc in the top quartile of the 1900 socioeconomic index within each county, as in Section 5.2. We use a random sample of four million records in the labor force with a recorded score. We fit a single jointly estimated BIRDIE model over the four blocs and the residual with complete pooling (McCartan et al., 2025). We again compare ℓ BISG against oracle ℓ BISG.

Figure A8 compares the estimated top-quartile share of each bloc at the county level against the truth. The top panel shows that the bias of ℓ BISG is close to that of oracle ℓ BISG: 4.3 against 1.9 ppts for Italian records, -2.6 against -3.6 for Scandinavian, -2.4 against

1.6 for Germanic, and -6.4 against -5.3 for West Slavic. The bottom panel shows that ℓ BISG is within two percentage points of county RMSE of oracle ℓ BISG for three of the four blocs: 11.0 against 9.4 ppts for Italian records, 7.1 against 7.7 for Scandinavian, and 15.7 against 13.1 for West Slavic. For Germanic records, the county RMSE is 13.8 against 5.8 ppts, reflecting the white Americans of native parentage carrying Germanic surnames, whose share among people carrying a list surname is the largest of the four blocs. The error is otherwise largest for West Slavic, the smallest bloc with the weakest list.

Overall, we find that ℓ BISG, which recovers the bloc geography from the coarse geography and the lists, approaches oracle ℓ BISG in accuracy, calibration, and downstream analysis validity after the additional step described above. The results depend on the lists being exclusive to their target groups; where they are not, as for the Germanic bloc, the downstream estimates are less accurate. This validation demonstrates the application of ℓ BISG to subgroups for which neither a surname table nor a geographic prior exists.

A7 Exact Prompts Used

All lists are generated by Gemini 3.0 Flash at temperature 1.0, with six calls for each prompt and 300 surnames requested per call (500 full names in Lebanon), ranked by the number of calls returning each surname. Every prompt substitutes a group description into a fixed skeleton.

A7.1 United States voter files

The skeleton is:

List $\{n\}$ common SURNAMES (last names only) of $\{description\}$ in the United States. Maximize diversity; avoid famous/celebrity surnames. One surname per line, UPPER-CASE, no first names, no numbering, no commentary.

with the following descriptions:

- White:
 - “non-Hispanic White American families of English and Irish descent”
 - “non-Hispanic White American families of German descent”
 - “non-Hispanic White American families of Italian, Polish, and other European descent”
- Black:
 - “African American / Black American families (common everyday surnames)”
 - “Black American families from the US South”
 - “Black American families (a diverse set, ordinary people not celebrities)”
- Hispanic:
 - “Mexican American families”

- “Cuban and Puerto Rican American families”
- “Central and South American (Salvadoran, Colombian, Dominican) families in the US”
- Asian:
 - “Chinese American families”
 - “Asian Indian American families”
 - “Vietnamese and Filipino American families”
 - “Korean and Japanese American families”

A7.2 Historical census

We use Gemini 3.0 Flash to write a meta prompt that asks the model to write the descriptions itself:

You are helping build surname lists for {origin} immigrants to the United States around 1900. Propose exactly three short descriptions of subgroups that together cover that immigrant population, distinguished by the region they came from or the region of America where they settled. Each description must complete the sentence ‘list 300 common surnames (last names only) of ...’. Return only the three descriptions, one per line, with no numbering.

The three descriptions returned for each origin are substituted into the skeleton:

List {n} common SURNAMES (last names only) of {description}. Maximize diversity; avoid famous/celebrity surnames. One surname per line, UPPERCASE, no first names, no numbering, no commentary.

Table A2 lists every description used. The list for the coarse white group, from which the residual class of Appendix A6 is formed, uses the same meta-prompt phrased for the white population of native parentage.

Origin	Descriptions
German	“German immigrant families in the rural US Midwest (ordinary farmers, not famous people)” “German immigrant families (Prussian, Bavarian, Saxon origins) in 19th-century America” “German-American families in the Great Plains and Wisconsin/Minnesota”
Irish	“Irish immigrant families in 19th-century America (ordinary, not politicians)” “Irish Catholic immigrant families in the US Midwest and East” “Irish Protestant immigrant families in the rural South and Appalachian regions”
English	“English immigrant families in 19th-century rural America” “English-American families (ordinary, not famous)”

Origin	Descriptions
	"English immigrant industrial workers in the urban Northeast and Midwest"
Canadian	"list 300 common surnames (last names only) of French-speaking immigrants from Quebec who settled in New England textile towns" "list 300 common surnames (last names only) of English-speaking immigrants from Ontario and the Maritimes who settled in the American Midwest and Northeast" "list 300 common surnames (last names only) of Icelandic and Scandinavian-descendant immigrants from the Canadian Prairies who settled in the Upper Midwest and Pacific Northwest"
Swedish	"Swedish immigrant families in the US Midwest (Minnesota, Illinois)" "Swedish-American families (ordinary 19th-century immigrants)" "Swedish immigrant workers and families in the Pacific Northwest and the Northeast."
Norwegian	"Norwegian immigrant families in the US Upper Midwest (Minnesota, the Dakotas)" "Norwegian-American farming families (ordinary, 19th century)" "Norwegian maritime and industrial laborers in the Pacific Northwest and coastal urban centers"
Italian	"Italian immigrant families in American cities around 1900 (ordinary labourers, not famous people)" "Southern Italian and Sicilian immigrant families in the United States" "Northern Italian immigrant families in the United States"
Polish	"Polish immigrant families in American industrial cities around 1900 (ordinary workers)" "Polish immigrant families in the US Midwest (Chicago, Milwaukee, Detroit)" "Polish farming families who settled in rural America in the 19th century"
Russian	"list 300 common surnames (last names only) of Ashkenazi Jewish families from the Pale of Settlement who entered through Ellis Island" "list 300 common surnames (last names only) of East Slavic peasants from Great Russian provinces who settled in industrial cities of the Northeast" "list 300 common surnames (last names only) of Volga German and Russian Mennonite farmers who moved to the Great Plains states"
Scottish	"list 300 common surnames (last names only) of Scottish immigrants from the Highlands and Islands who settled in the rural Southern United States" "list 300 common surnames (last names only) of Lowland Scots who emigrated to industrial urban centers in the American Northeast and Midwest" "list 300 common surnames (last names only) of Ulster Scots who arrived via Northern Ireland to settle along the Appalachian frontier and Piedmont regions"

Origin	Descriptions
Austrian	"list 300 common surnames (last names only) of German-speaking immigrants from the Austrian heartland and Alpine provinces" "list 300 common surnames (last names only) of Polish and Ukrainian immigrants from the Crown Land of Galicia" "list 300 common surnames (last names only) of Czech and Slovak immigrants from Bohemia, Moravia, and Upper Hungary"
Czech	"Bohemian and Moravian immigrant families in the United States around 1900" "Czech immigrant farming families in Nebraska, Iowa and Texas" "Czech industrial and mining laborers in the Upper Midwest and Mid-Atlantic states around 1900"
Danish	"list 300 common surnames (last names only) of Danish immigrants from Jutland who settled in the rural Midwest" "list 300 common surnames (last names only) of Danish immigrants from the islands of Zealand and Funen" "list 300 common surnames (last names only) of Danish converts who emigrated to Utah and the Intermountain West"
French	"list 300 common surnames (last names only) of French Canadian immigrants who settled in New England textile towns" "list 300 common surnames (last names only) of European French immigrants who arrived through the Port of New York" "list 300 common surnames (last names only) of Basque and southwestern French immigrants who settled in the American West"
Swiss	"list 300 common surnames (last names only) of Swiss German immigrants who settled in the agricultural communities of the American Midwest" "list 300 common surnames (last names only) of French-speaking immigrants from the Romandy region of Switzerland" "list 300 common surnames (last names only) of Italian-speaking Swiss immigrants from the canton of Ticino who settled in California"
Welsh	"list 300 common surnames (last names only) of Welsh coal miners and steelworkers who settled in the industrial communities of Pennsylvania and Ohio" "list 300 common surnames (last names only) of Welsh agricultural families and livestock farmers who migrated to the rural Midwest and Great Plains" "list 300 common surnames (last names only) of Welsh slate quarrymen from North Wales who established settlements in New England and upstate New York"
Dutch	"Dutch immigrant families in the US Midwest (Michigan, Iowa) in the 19th century" "Dutch-American farming families (ordinary, not famous)" "Dutch immigrant families who settled in New York and New Jersey urban centers during the late 19th and early 20th centuries"

Origin	Descriptions
Hungarian	"Hungarian immigrant families in American industrial towns around 1900" "Magyar immigrant families in the United States (ordinary workers, not famous people)" "Hungarian immigrant families who settled in agricultural communities and rural farming regions of the American Midwest around 1900"
Finnish	"Finnish immigrant families in the US Upper Midwest (Michigan, Minnesota) around 1900" "Finnish immigrant mining and farming families in the United States" "Finnish immigrant communities in the Pacific Northwest and coastal Massachusetts around 1900"
Portuguese	"list 300 common surnames (last names only) of Azorean immigrants who settled in the fishing and textile communities of New England" "list 300 common surnames (last names only) of Madeiran immigrants who settled in the agricultural regions of California and Illinois" "list 300 common surnames (last names only) of Continental Portuguese immigrants who settled in the industrial urban centers of New Jersey and New York"
Belgian	"list 300 common surnames (last names only) of Flemish agricultural settlers in the American Midwest" "list 300 common surnames (last names only) of Walloon industrial workers in Pennsylvania and West Virginia" "list 300 common surnames (last names only) of Belgian immigrants from the border provinces of East and West Flanders"
White	"list 300 common surnames (last names only) of white families of native parentage in the Northeast and Mid-Atlantic states" "list 300 common surnames (last names only) of white families of native parentage in the Southern United States" "list 300 common surnames (last names only) of white families of native parentage in the Midwest and Western states"

Table A2: Model-written descriptions for the historical census lists.

A7.3 Lebanon

Each prompt embeds an English group description and four to five example full names, held fixed within the sect, into an Arabic skeleton. The skeleton renders in English as:

Write {n} full names of people from {descriptor}. Examples: {examples}. Avoid the family names of famous politicians and celebrated families. Write varied, ordinary family names. Write exactly one name per line, with no numbers, no explanation, and no commentary. (Rendered from the Arabic; the original prompts appear in the replication archive.)

The example names are ordinary full names of the sect. Table A3 lists every description

used.

Table A3: Group descriptions used in the Lebanese sect prompts.

Prompt stem	Completions
Lebanese Shia Muslim families from [x]	southern villages (Nabatieh, Tyre rural areas — ordinary families, NOT famous religious or political figures) Baalbek-Hermel and the northern Beqaa (ordinary villagers) the Beirut southern suburbs (Dahieh) — everyday families
Lebanese Shia Muslim families from [x] (ordinary families, not famous people)	the Nabatieh and Bint Jbeil districts Beirut's southern suburbs the Baalbek-Hermel countryside the Tyre and Sarafand coast the Marjeyoun and Khiam area the Jezzine and Iqlim al-Tuffah villages the Bekaa villages around Hermel the Shia families of Baalbek city the villages of the Bint Jbeil highlands the Sidon and Zahrani hinterland the Beirut families of Chiyah and Ghobeiry the southern border villages near Aitaroun and Aita al-Shaab the villages of Kfar Roummene and Zawtar the Shia families of Baalbek's Sharawneh quarter
Lebanese Sunni Muslim families from [x]	rural Akkar district (northern Lebanon, ordinary villagers — NOT politicians or famous figures) Tripoli and the northern coastal region (ordinary residents) Beirut and the Beqaa valley (everyday families, not political dynasties)
Lebanese Sunni Muslim families from [x] (ordinary families, not famous people)	the city of Tripoli and its old quarters Beirut's Tariq al-Jadida and Basta the Iqlim al-Kharrub and Sidon countryside the western Bekaa and Rashaya the Akkar plain and Halba the Minieh-Dinnieh district the old Sidon families the Arqoub and Shebaa area

Prompt stem	Completions
	the Bekaa town of Majdal Anjar and Saadnayel the coastal villages between Batroun and Chekka the Beirut families of Mazraa and Ras al-Nabaa the Hermel and northern Bekaa Sunni villages
Lebanese Maronite Christian families from [x]	the Kesrouan and Metn districts in Mount Lebanon (ordinary families) Bsharri, Zgharta, and the northern mountain region the Chouf and southern Mount Lebanon (everyday families)
Lebanese Maronite Christian families from [x] (ordinary families, not famous people)	the Bsharri and Qadisha valley villages the Kesrouan and Jbeil hills the Metn and upper Matn villages the Zgharta and Koura districts the Batroun highlands the Jezzine and southern Maronite villages the Baabda and Aley Christian villages the Ehden and Zawiya families the Aakoura and Laqlouq mountain villages the Zahle and Bekaa Maronite families the Ftouh Kesrouan coastal towns the Chouf Maronite villages the Tannourine and Hadath al-Jebbeh villages the Kfarsghab and Hasroun families the Kartaba and Mayfouq villages the Ghosta and Harissa families the Bikfaya and Dhour Choueir families the southern villages of Ain Ebel and Rmeish the Deir al-Ahmar and Baalbek Maronite families the Douma and Tannourine highland families
Lebanese Greek (Roman) Orthodox Christian families from [x]	the Koura district in northern Lebanon (ordinary families, NOT famous politicians) Beirut's Achrafieh and Ras Beirut quarters Tripoli, El Mina, and surrounding northern coastal areas
Lebanese Greek Orthodox Christian families from [x] (ordinary families, not famous people)	the Koura district villages Beirut's Achrafieh and Rmeil the Marjeyoun and Hasbaya area the Zahle and central Bekaa the city of Tripoli's Orthodox families

Prompt stem	Completions
	the Amioun and Kfaraakka villages the Metn Orthodox villages the Akkar Orthodox villages the Shwayfat and Aley Orthodox families the Beirut families of Mousaitbeh and Ras Beirut the Batroun and Hamat coastal families the Rashaya and Wadi al-Taym Orthodox villages the villages of Kousba and Bterram the Anfeh and Ras Masqa coastal villages the Beit Mery and Baabdat families
Lebanese Druze families from [x]	the Chouf district villages (ordinary villagers — NOT the Jumblatt, Arslan, or Hamadeh political families, NOT famous figures, ordinary common surnames) Aley and Baabda mountain villages (everyday families) the Hasbaya and Rashaya districts (southeastern Lebanon)
Lebanese Druze families from [x] (ordinary families, not famous people)	the Chouf mountain villages the Aley and Upper Metn district the Hasbaya and Rashaya area the Wadi al-Taym villages the Baakline and Deir al-Qamar families the Shwayfat and Choueifat area the Hermon foothill villages the Aley district towns of Bhamdoun and Souk al-Gharb the Iqlim al-Kharrub Druze villages the Mount Lebanon villages of Ain Dara the Metn Druze families the Rashaya al-Wadi villages the Mukhtara and Barouk villages the Ain Zhalta and Niha families the Abey and Baysour villages the Ras al-Metn and Salima families the Kfarmatta and Aitat villages the Btater and Kayfoun families the Jahliyah and Semqaniyah villages the Aramoun and Bennay families the Hasbaya villages of Shwayya and Miness

Prompt stem	Completions
Lebanese Melkite Greek Catholic (Roman Catholic) families from [x]	the Zahle area in the Beqaa valley (ordinary families)
	the Metn district and Beirut
	southern Lebanon (Jezzine area) and other regions
Lebanese Greek Catholic (Melkite) families from [x] (ordinary families, not famous people)	the Zahle and Furzol area
	the Sidon and Jezzine hinterland
	the Beirut Melkite families
	the Rashaya and western Bekaa
	the Marjeyoun and Deir Mimas villages
	the Baalbek Melkite families
	the Chouf Melkite villages
	the Tyre and southern Melkite families
	the Metn Melkite families
	the Aley and Bhamdoun Melkite families
	the Baabda district Melkite families
	the Bekaa villages around Rayak and Talia
	the Zahle quarters of Maalaka and Karak Nuh
	the Jezzine villages of Roum and Kfarhouana
	the Ras Baalbek and Qaa Melkite families
“Lebanese Armenian Orthodox families living in Bourj Hammoud and Beirut. Surnames are Armenian in origin but spelled in Arabic letters as used in Lebanese voter registries. Provide diverse Armenian family names (NOT just famous figures), and include common Armenian surname suffixes -ian and -oun and -jian rendered in Arabic”	
Lebanese Armenian Orthodox families from [x]	the Antelias and Anjar areas (Armenian-origin surnames in Arabic script)
“Lebanese Armenian Orthodox families with surnames ending in the typical Armenian patronymic suffixes (-yan, -ian) transliterated to Arabic script”	
Lebanese Armenian Apostolic families from [x] (ordinary families, not famous people)	the Bourj Hammoud quarter
	the Anjar settlement in the Bekaa
	Beirut’s Nor Marash and Nor Sis quarters
	the Antelias and Metn Armenian families
	the Zalka and Jdeideh Armenian families
	the Tripoli Armenian community
	the Ainjar and Bekaa Armenian villages
	the Armenian families of Dora and Nabaa
	the Armenian families descended from Cilicia and Adana

Prompt stem	Completions
	the Armenian families descended from Marash and Aintab
	the Armenian families descended from Urfa and Kharpert
	the Bourj Hammoud districts of Sanjak and Arax

A8 Additional Empirical Results

A8.1 Census-based name list generation

τ	Shared	White		Black		Hispanic		Asian	
		Cov.	Prec.	Cov.	Prec.	Cov.	Prec.	Cov.	Prec.
0.20	7.0%	0.28	0.79	0.51	0.37	0.50	0.82	0.42	0.60
0.30	0.04%	0.28	0.79	0.33	0.46	0.50	0.82	0.42	0.64
0.40	0.00%	0.28	0.79	0.17	0.62	0.50	0.82	0.41	0.65
0.50	0.00%	0.28	0.79	0.11	0.75	0.50	0.83	0.40	0.73
0.60	0.00%	0.28	0.79	0.08	0.85	0.49	0.83	0.39	0.74

Table A4: Sensitivity of the Census-based lists to the probability floor τ , where a surname enters the list for group r when $\mathbb{P}(R = r \mid S = s) \geq \tau$. Coverage and precision are defined as in Table 1, computed on the Florida file. “Shared” is the share of registrants carrying a surname on more than one of the four lists. The floor used in the paper is 0.30.

Table A4 reports the Census lists under values of τ ranging from 0.20 to 0.60. The White, Hispanic, and Asian lists are largely unchanged. For these groups, the top 1,000 surnames by the ranking rule clear even the highest value of τ . The Black list is far more sensitive, clearly trading off coverage for precision as τ rises. τ also determines the extent to which the group lists have overlapping surnames: at $\tau = 0.20$, 7.0% of registrants have a surname that belongs to two groups, while at $\tau = 0.30$, that share falls to 0.04%, and above $\tau = 0.30$, it is zero. We therefore use $\tau = 0.30$, the smallest measured floor at which the lists are close to disjoint.

A8.2 Robustness of the LLM lists

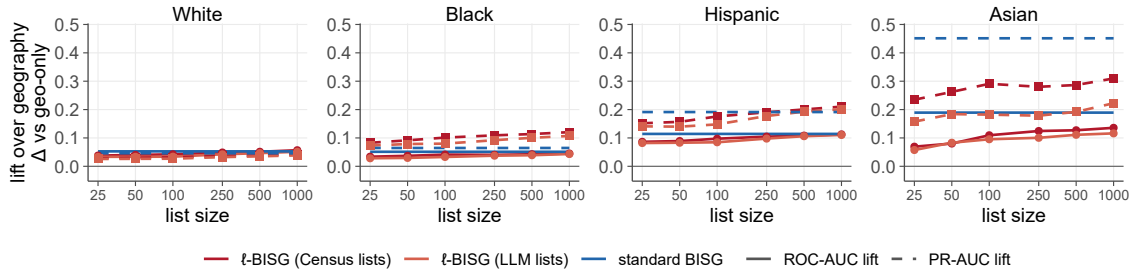


Figure A9: Discrimination lift over geography alone, in ROC-AUC and PR-AUC, against the number of surnames per group, on the Florida voter file with leave-one-out tract shares as the geographic baseline, for the Census and LLM lists of Table 1, with standard BISG drawn as a flat reference.

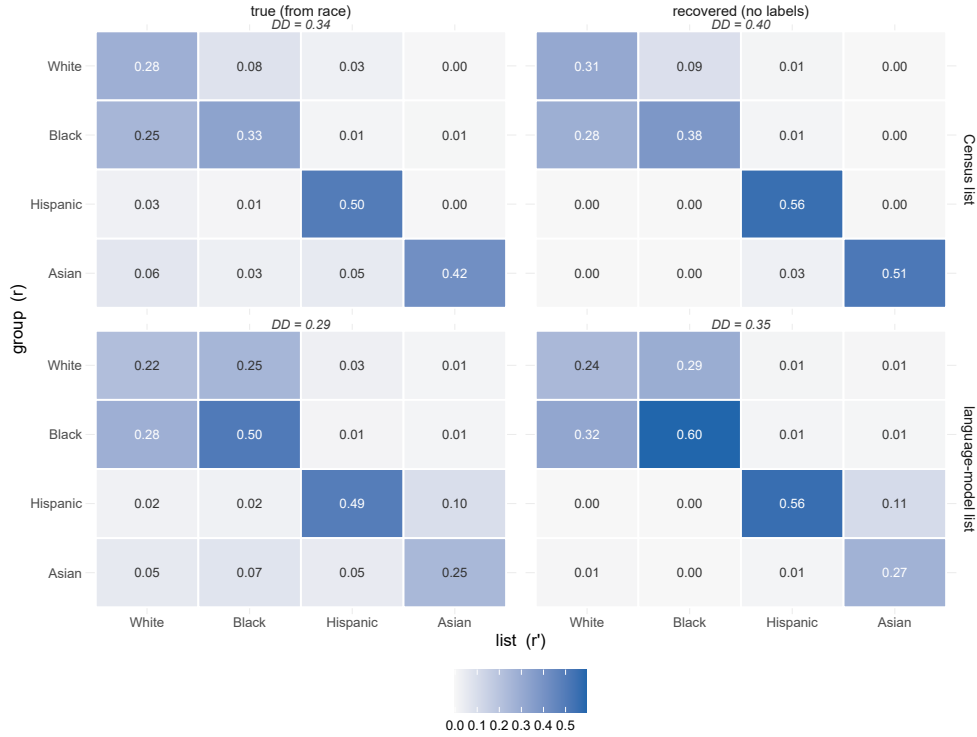


Figure A10: List quality on the Florida voter file. Each cell is $\pi_{r,r'}$, the rate at which members of group r (rows) carry a name on list r' (columns), for the Census lists (top) and the LLM lists (bottom). The left column computes Π from the self-reported race (i.e., ground-truth). The right column estimates Π from the tract-level list hit rates and the published tract race shares, with no individual race labels. The number above each panel is $\gamma(\Pi)$.

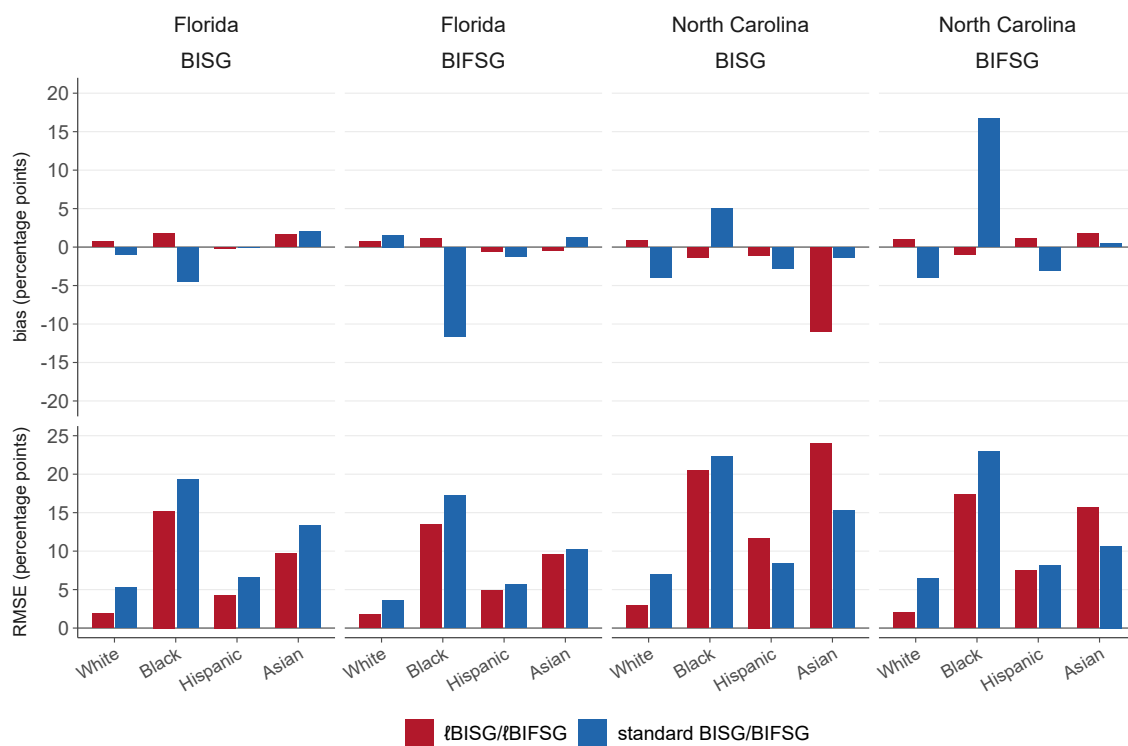


Figure A11: County-level race-specific estimates of the Democratic share of registrants on the Census lists, otherwise as Figure 4.